TechPrime AI

System Overview and Role Functions

TechPrime AI: A Barcode-Integrated Intelligent Ordering and Predictive Inventory Management System with Machine Learning-Based Product Recommendation for EasyPC One Oasis Branch

1. About the System

TechPrime AI is a **barcode-integrated intelligent ordering and predictive inventory management system** with machine learning-based product recommendation, developed for **EasyPC One Oasis Branch** (Rosario, Pasig City). It brings ordering, inventory control, customer assistance and decision support into one web platform with a single shared database, so every barcode scan, walk-in sale and online order updates stock in real time.

To support decision-making and customer service, it adds an **XGBoost** model that forecasts product demand and sales from historical POS records, an **SVM-based NLP chatbot (Primo)** that understands and answers customer inquiries (with a language model that phrases its shop-related replies), and a **Build a PC (Tech & Match)** tool that checks component compatibility and recommends matching parts.

2. Main Functions

- **Barcode-Integrated Inventory:** product catalog, printable barcode labels, scan-based stock-in, and real-time low / critical / out-of-stock alerts.
- **Point of Sale (POS):** barcode checkout for walk-in customers (Cash, GCash, Maya, Card) with VAT-inclusive receipts; each sale deducts stock immediately.
- **Online Ordering:** browse, search, wishlist, cart and checkout (Lalamove delivery or store pick-up), order tracking and product reviews.
- **Forecasting and Reports (XGBoost):** 1–3 month product demand and sales forecasts from historical POS sales, restocking alerts, and sales reports that can be printed or exported to CSV, Excel and PDF.
- **Product Recommendation (Build a PC):** step-by-step Tech & Match builder with compatibility checks, build score, power estimate and saved builds.
- **Primo AI Chatbot (SVM):** NLP and LLM assistant that classifies customer questions by intent and answers product, stock, order and build inquiries from the live catalog. A language model phrases shop-related replies; off-topic questions are politely declined.
- **Security and Audit:** role-based access, 15-minute idle timeout, login lockout, password rules, CSRF protection and searchable, paginated activity logs.
- **Staff Messaging:** internal chat with attachments among the Admin, Retail Officers and Inventory Custodians.

3. User Roles at a Glance

Role	Main Responsibility	Modules
Admin	Oversees users, security, forecasts and the audit trail	Dashboard, Forecast (Product Demand, Sales), Messages, Manage Users, Activity Logs, My Profile, Settings
Retail Officer	Monitors sales performance and forecasts	Dashboard, History, Forecast, Messages, Profile

Role	Main Responsibility	Modules
Inventory Custodian	Manages products, receiving, online orders and reviews	Dashboard, Orders, Stocks (Inventory, Add New Product), Forecast, Reviews, Messages, Activity, Profile
Cashier	Processes walk-in sales at the counter	Dashboard, POS Checkout, Receipt, Stock Alerts, Profile
Client	Shops online and plans PC builds	Home, Primo, Shop, Category, Product Details, Build a PC, Saved Builds, Wishlist, Cart, Checkout, My Account, Product Review, Account Settings, Notifications, Privacy Policy

4. Admin

The Admin oversees the whole system: user accounts, security policy, forecasts and the system-wide audit trail.

4.1 Dashboard

Store-wide summary for the current month.

- Totals for clients, retail officers and inventory custodians
- Monthly revenue, orders and completed deliveries
- Restocking Alerts: products at or below the low-stock threshold
- Print button with a print-ready layout (logo, title, date range and prepared-by)

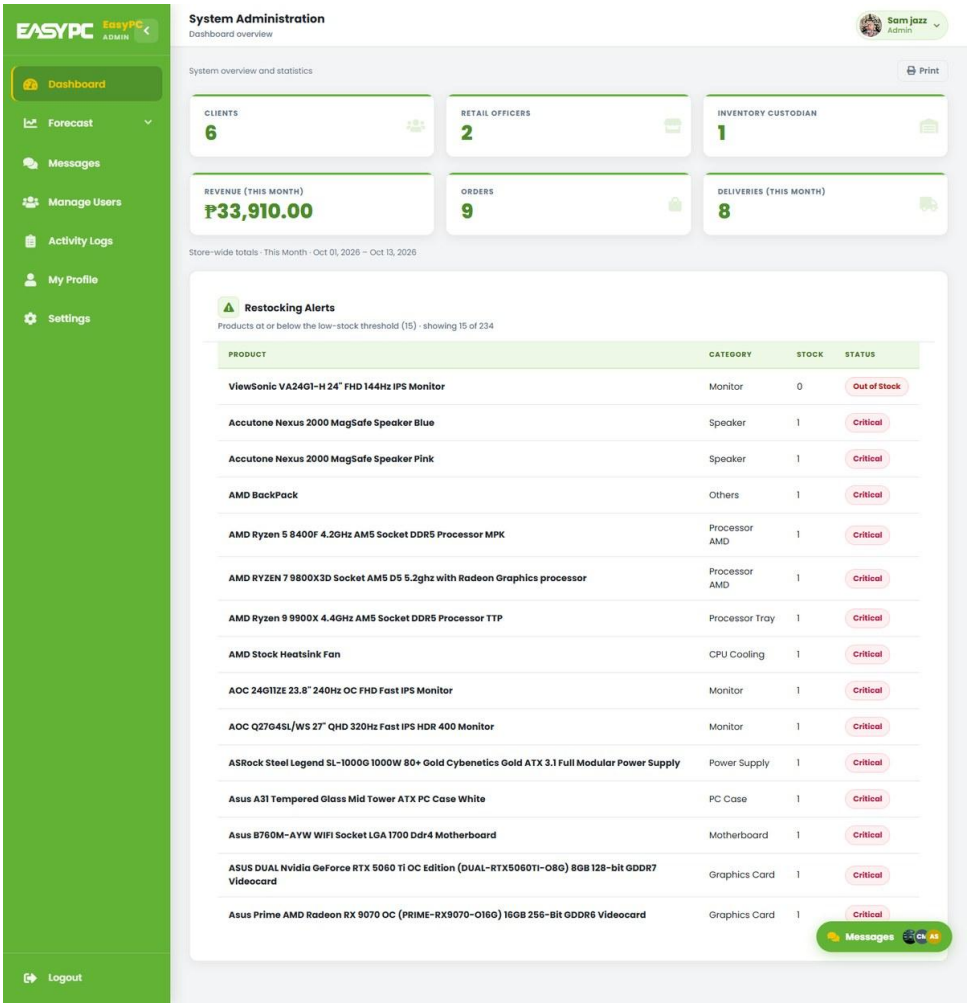


Figure 1. Admin: Dashboard

4.2 Forecast

Projects product demand and sales 1–3 months ahead. The Forecast menu has two pages: Product Demand and Sales.

- Product Demand: units sold, forecast units, likely range, forecast revenue, trend and confidence per product
- Sales: actual vs. forecast monthly sales chart with the projected total for the next 3 months
- Filter by category and period; switch between chart and table view

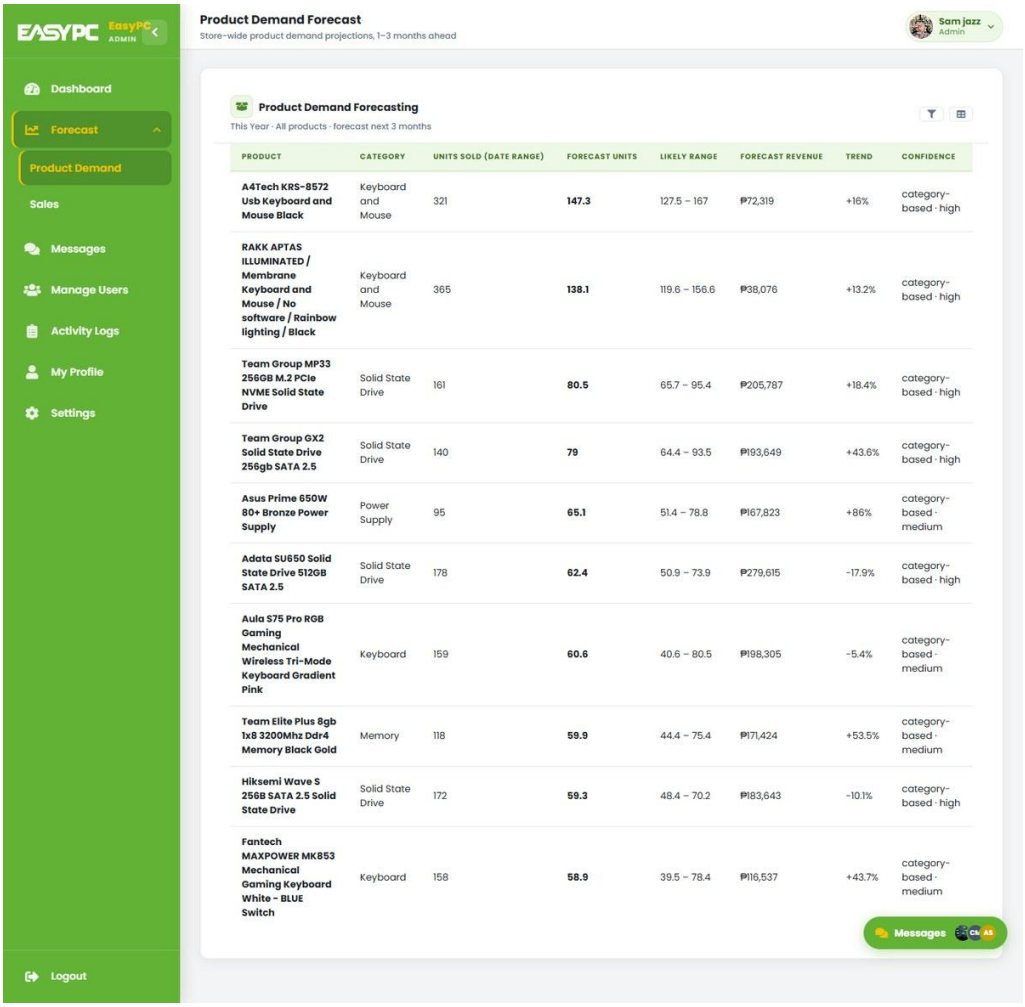


Figure 2. Admin: Forecast (Product Demand)

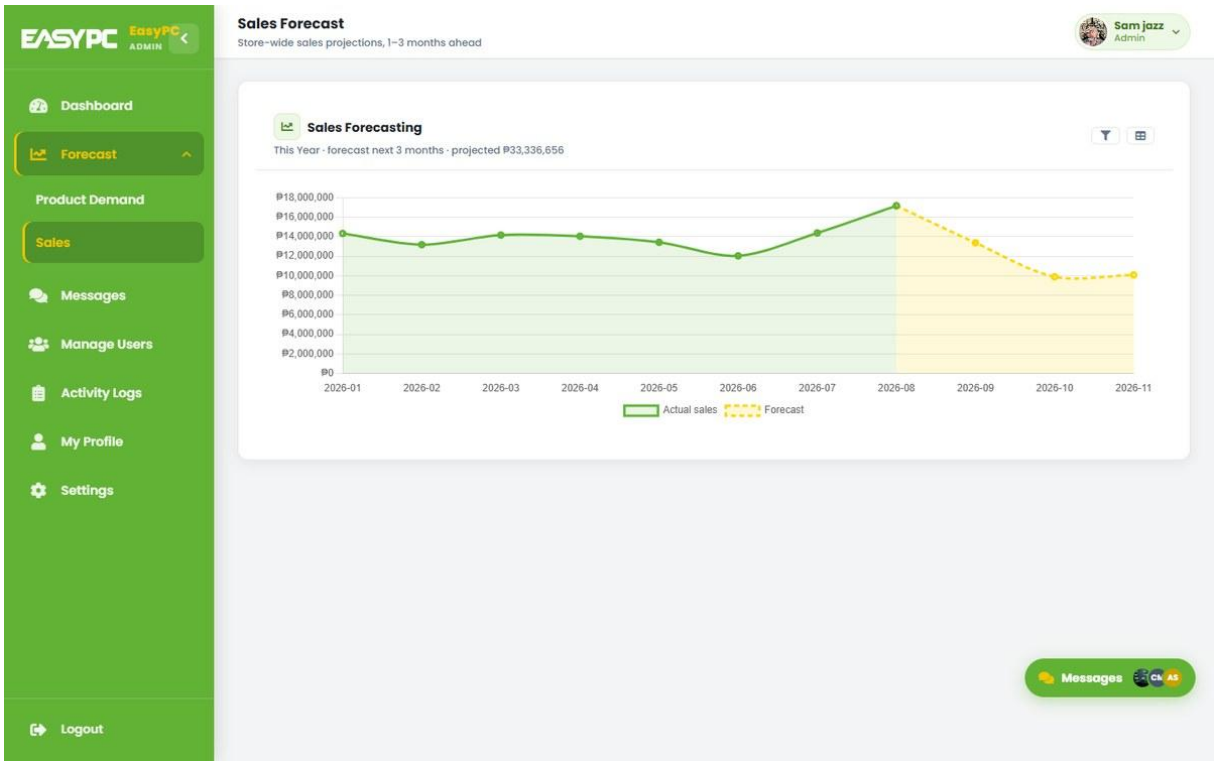


Figure 3. Admin: Forecast (Sales)

4.3 Messages

Internal staff chat with Retail Officers and Inventory Custodians.

- Search staff and open a conversation
- Send messages and attachments; react, forward, pin or unsend
- Opens with the latest 10 messages; older messages load 10 at a time as you scroll up
- Also available as a floating Messages widget on every staff page

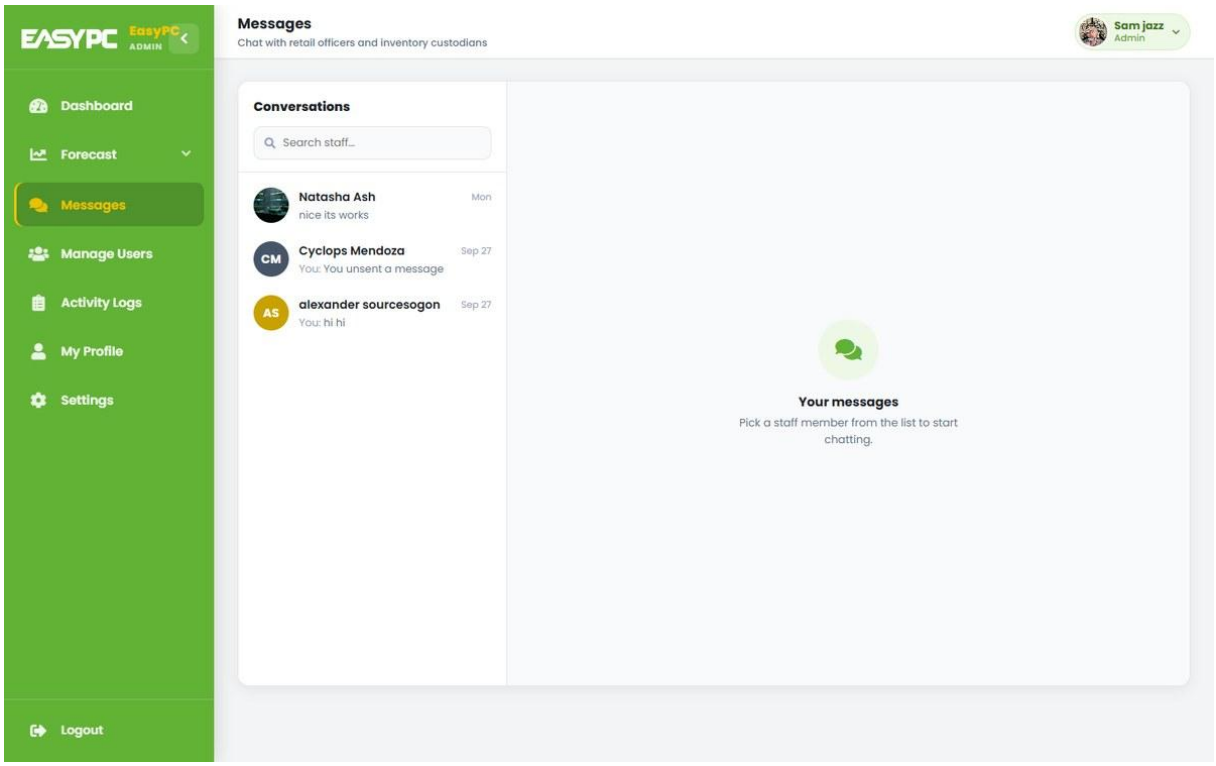


Figure 4. Admin: Messages

4.4 Manage Users

View and control every user account by role.

- Filter accounts by role; see verification and account status
- Create Staff Account for Retail Officers, Inventory Custodians and Cashiers
- Block, unblock or delete accounts

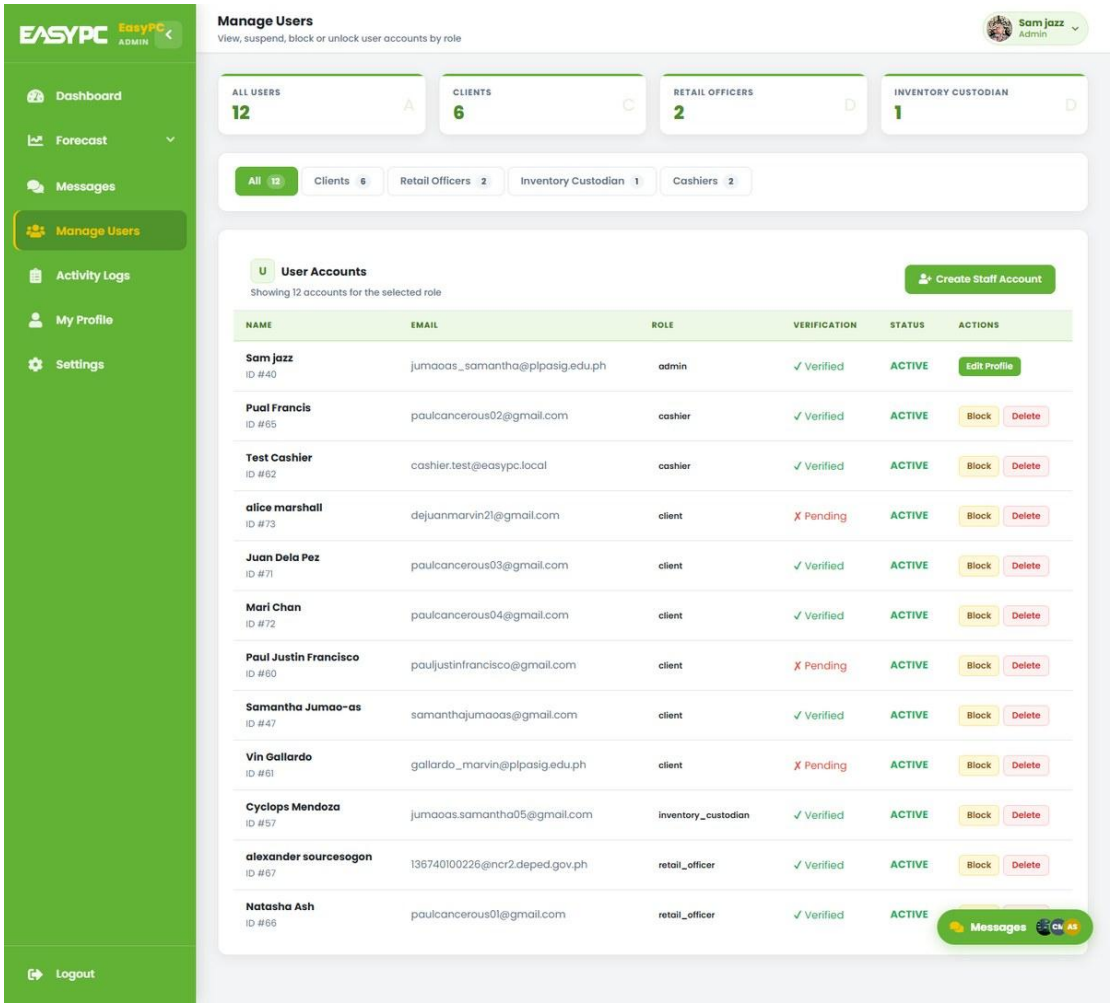


Figure 5. Admin: Manage Users

4.5 Activity Logs

System-wide audit trail of significant actions.

- Shows timestamp, user, role, action, details and IP address
- Covers logins, page views, profile changes, orders and stock updates
- Search across all logs by user, email, action, details or IP address
- 25, 50 or 100 rows per page, with page navigation

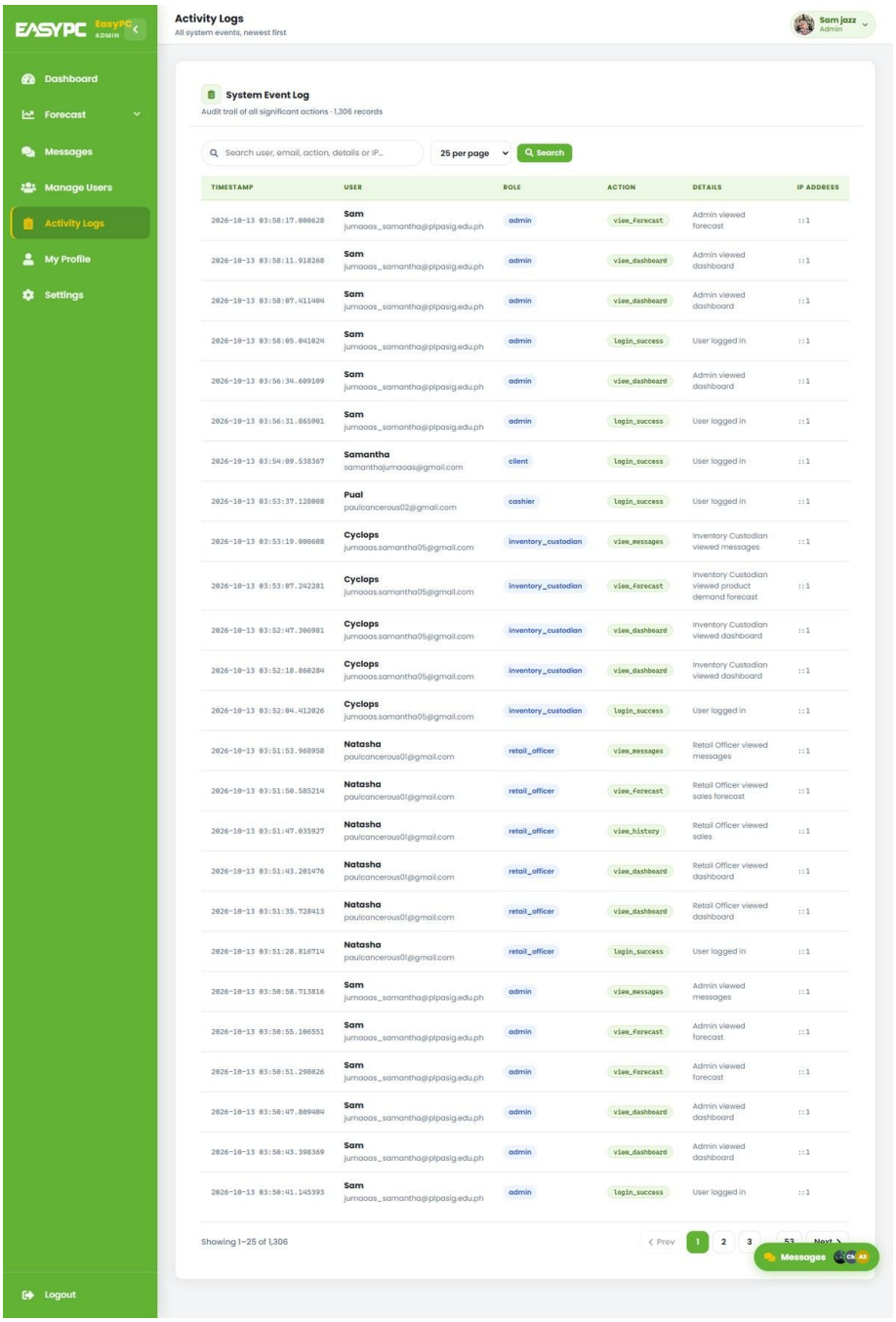


Figure 6. Admin: Activity Logs

4.6 My Profile

Lets the user update their own account details.

- Edit name, age and address; change the profile picture
- Change password with a live checklist of the password rules

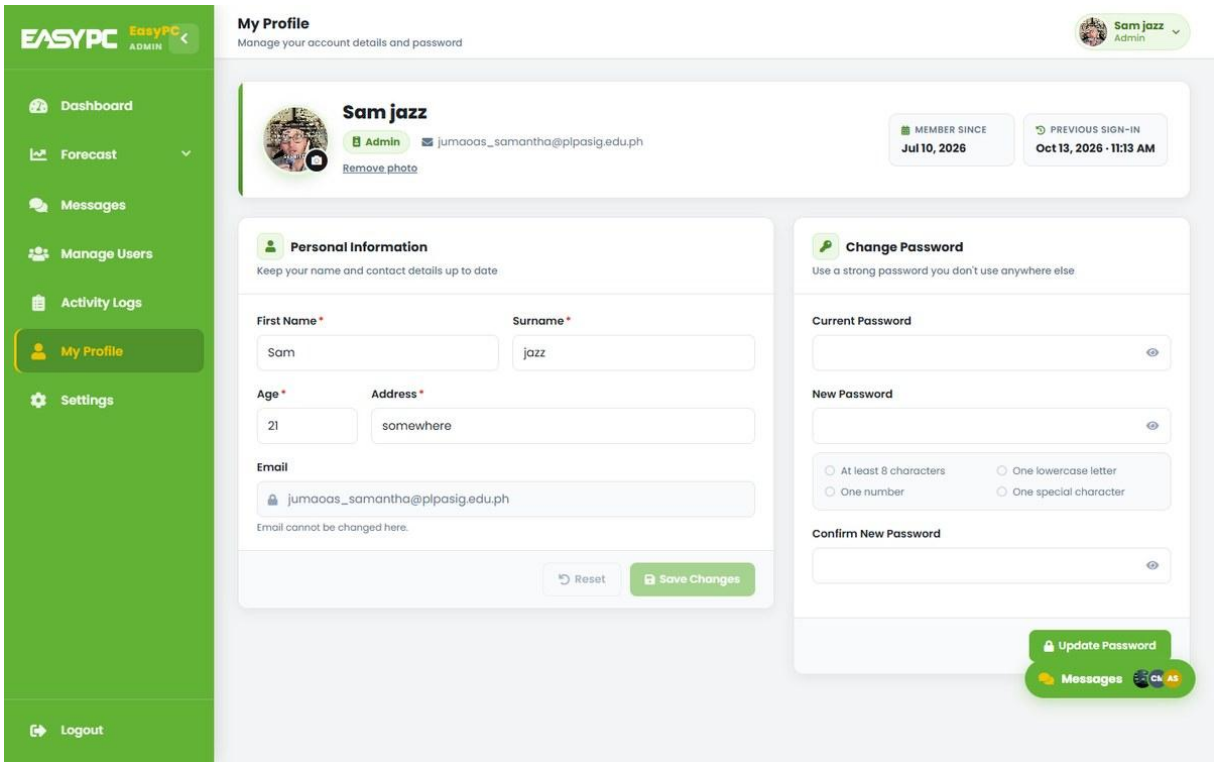


Figure 7. Admin: My Profile

4.7 Settings

Sets the login and password security policy.

- Login Security: failed attempts allowed before an account locks (1–10); applies immediately
- Password Complexity: minimum length, uppercase, lowercase, number, special character; applies to new passwords
- Live Preview of the active password rules

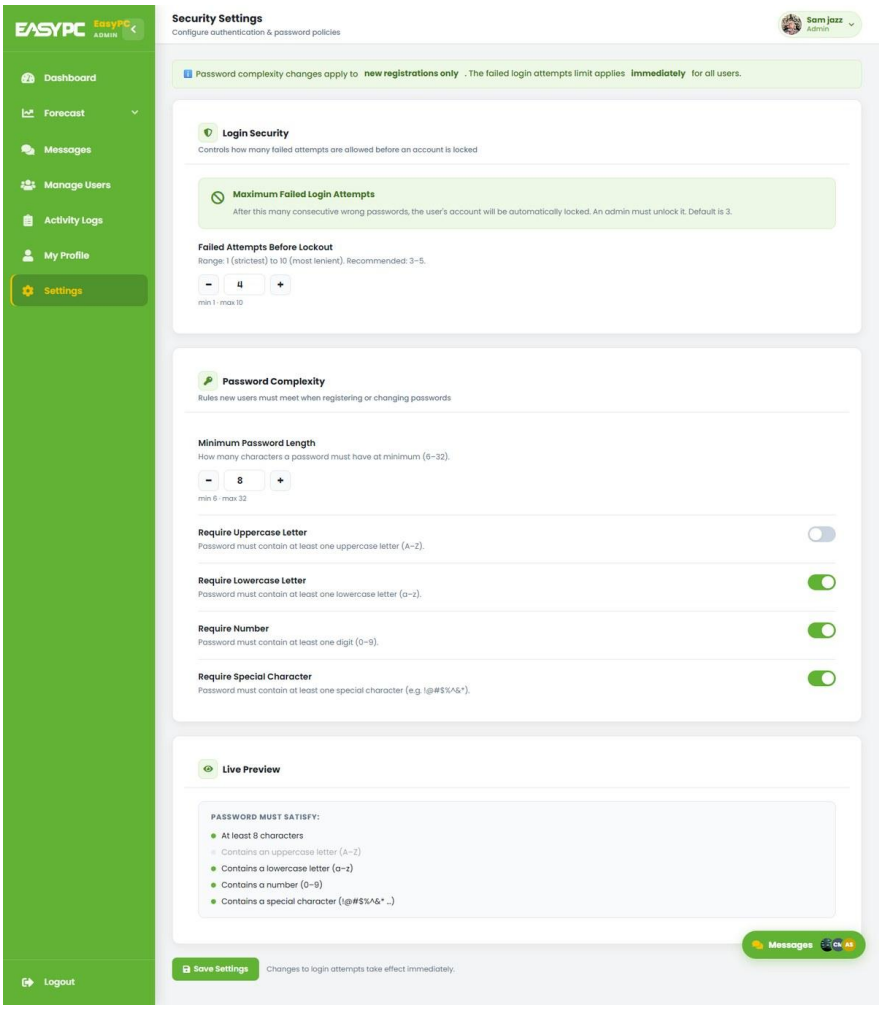


Figure 8. Admin: Settings

5. Retail Officer

The Retail Officer monitors sales performance and forecasts to guide restocking and sales decisions.

5.1 Dashboard

Sales analytics on one page.

- This month's total sales, transactions, units sold and average transaction
- Monthly Sales Report for the last 12 months, in table or graph view
- Sales by Category chart, filtered by period and category
- Print, Export CSV and Export PDF (a downloadable PDF report with the summary, monthly sales and sales by category)

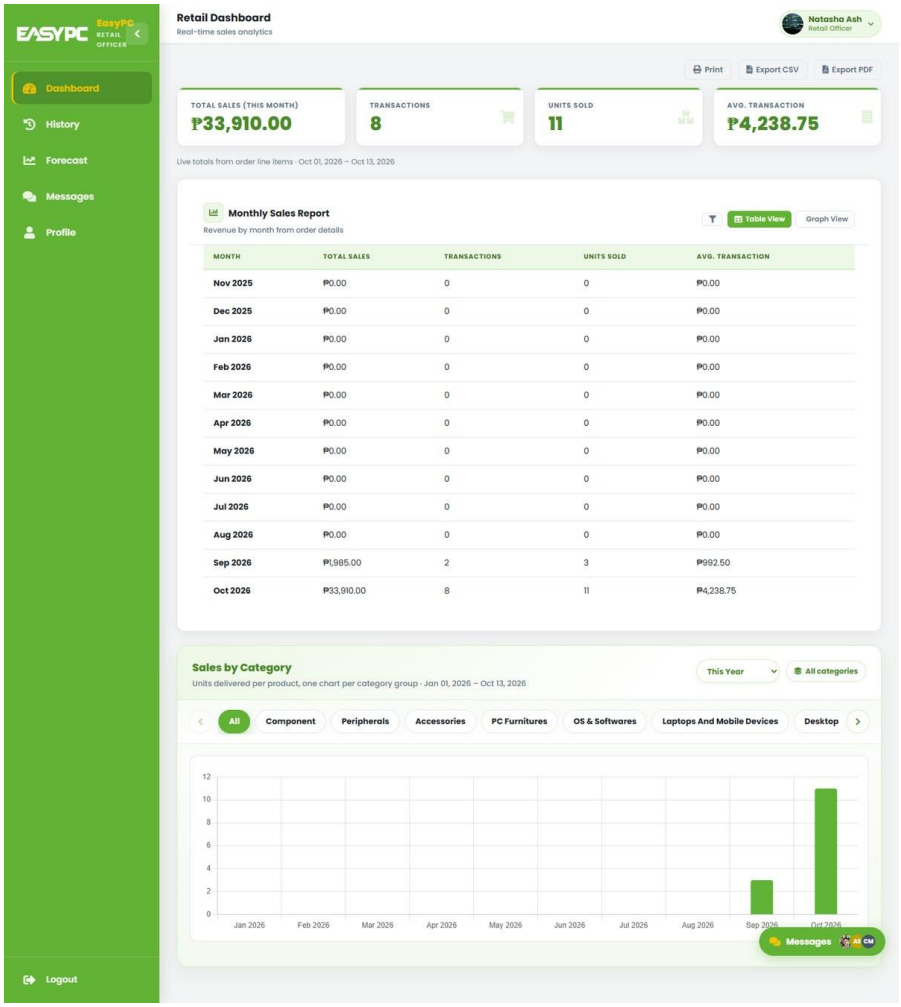


Figure 9. Retail Officer: Dashboard

5.2 History

Sales History of completed deliveries.

- Total deliveries, shipments in range and success rate
- Completed Deliveries table (shipment, order, customer, carrier, total, delivered date); click a row for its items
- Filter by period or a custom date range
- Print, Export CSV, Export Excel (.xlsx with Summary, Deliveries and Items sheets) and Export PDF

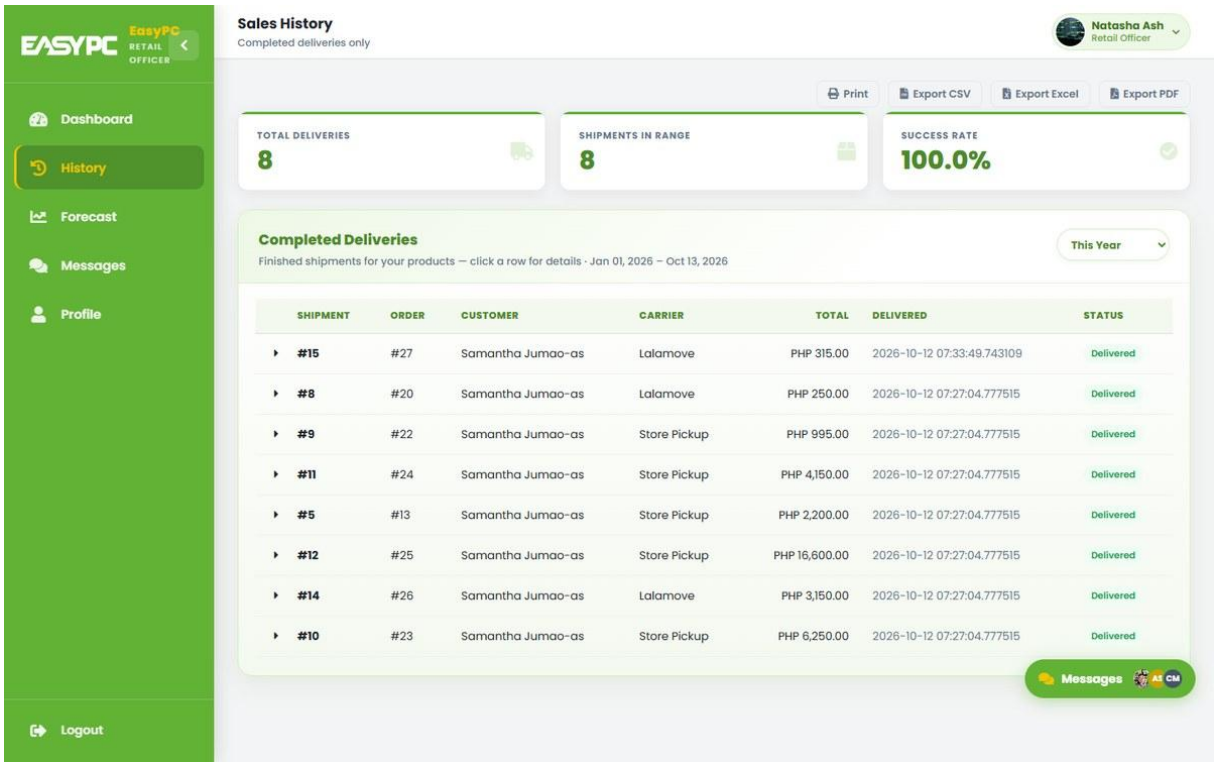


Figure 10. Retail Officer: History

5.3 Forecast

Sales Forecast for the next 1–3 months, based on historical sales.

- Actual vs. forecast monthly sales chart with the projected total
- Table view with month, units, projected sales (PHP) and likely range
- Filter by year, period and product category

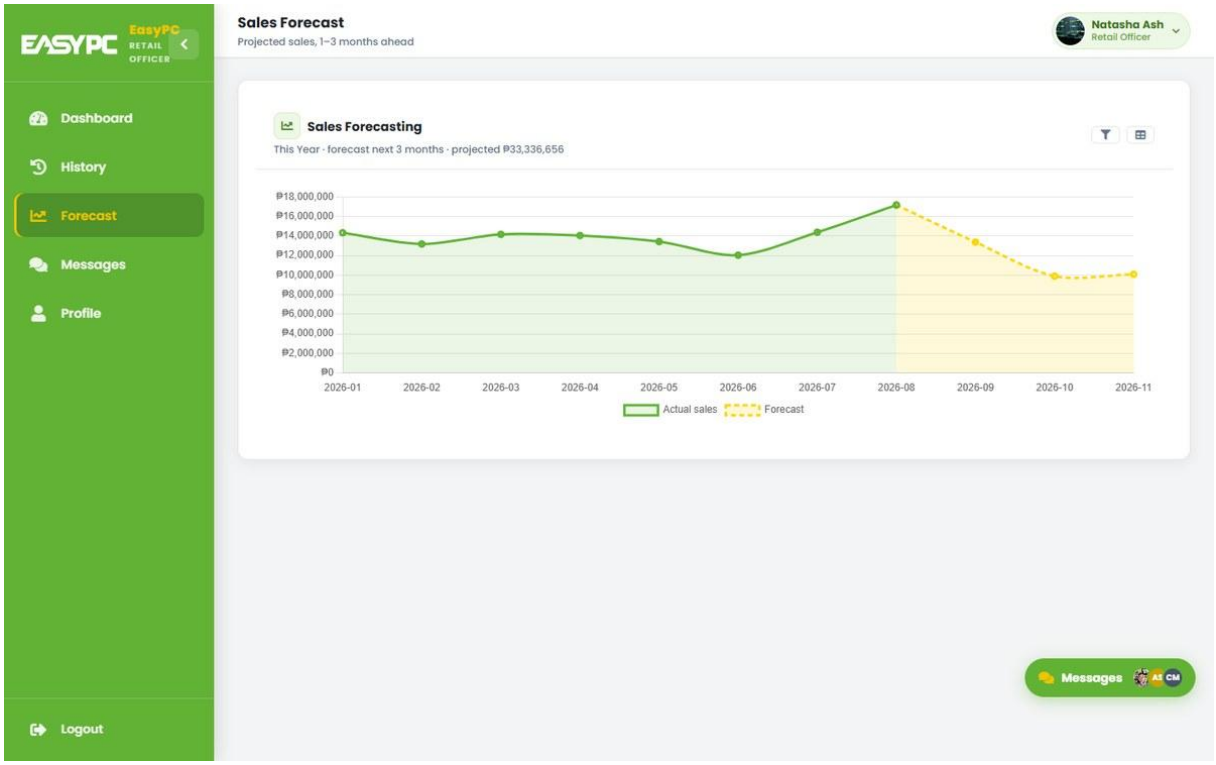


Figure 11. Retail Officer: Forecast

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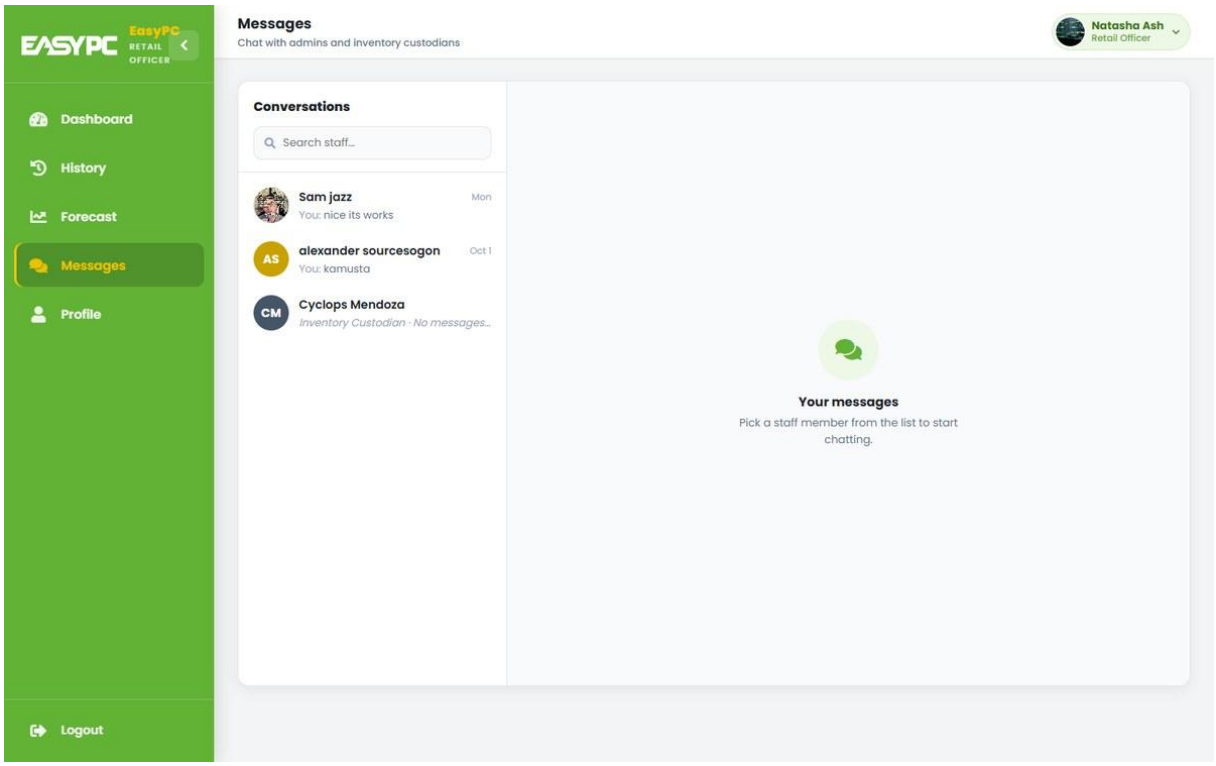


Figure 12. Retail Officer: Messages

5.5 Profile

Lets the user update their own account details.

- Edit name, age and address; change the profile picture
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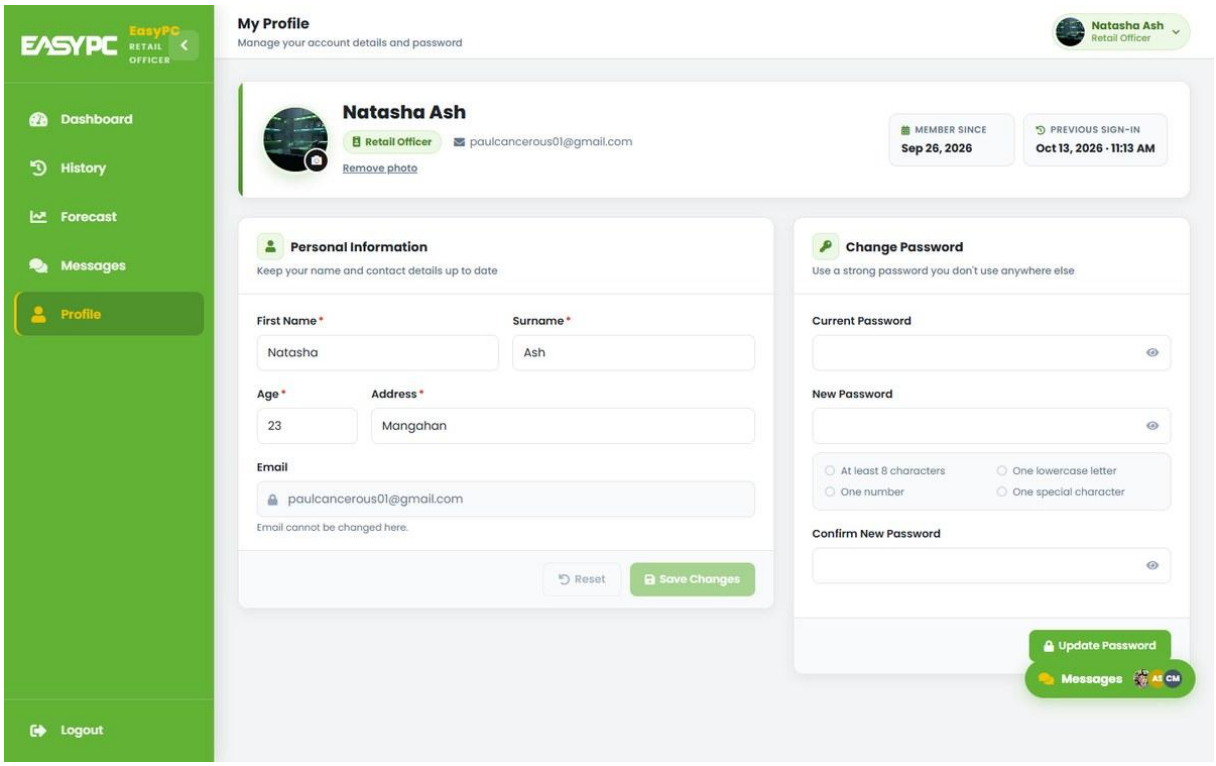


Figure 13. Retail Officer: Profile

6. Inventory Custodian

The Inventory Custodian manages the product catalog, receives deliveries, processes online orders and answers product reviews.

6.1 Dashboard

Overview of deliveries and stock levels.

- Total and completed deliveries, pending shipments and catalog size
- Total stock units, low-stock and out-of-stock counts
- Products Stocked Per Month chart, filtered by year and category
- Print button with a print-ready layout

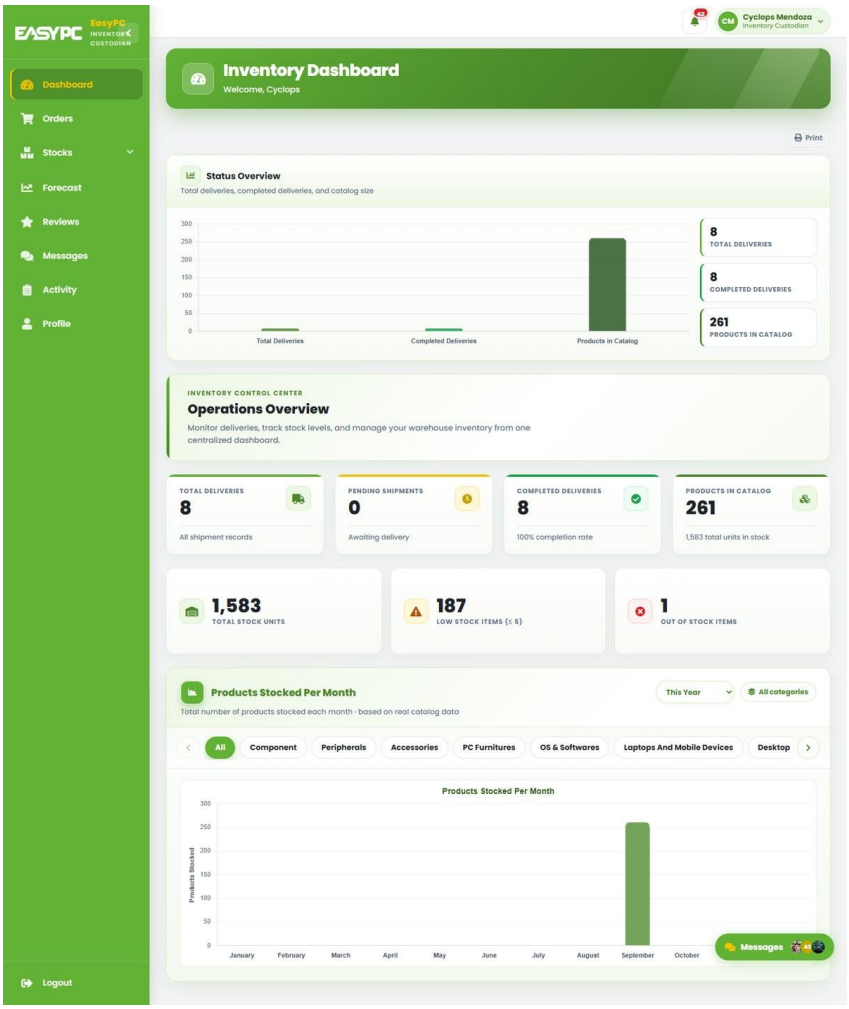


Figure 14. Inventory Custodian: Dashboard

6.2 Orders

Handles customer online orders.

- Accept new orders and move them to the next status (e.g. To Ship, Completed); the client is notified
- Bulk update: select several orders with the same status and update them together
- Filter by status; view customer, items, total and delivery or pick-up details
- Weekly Total Deliveries chart for the last 8 weeks

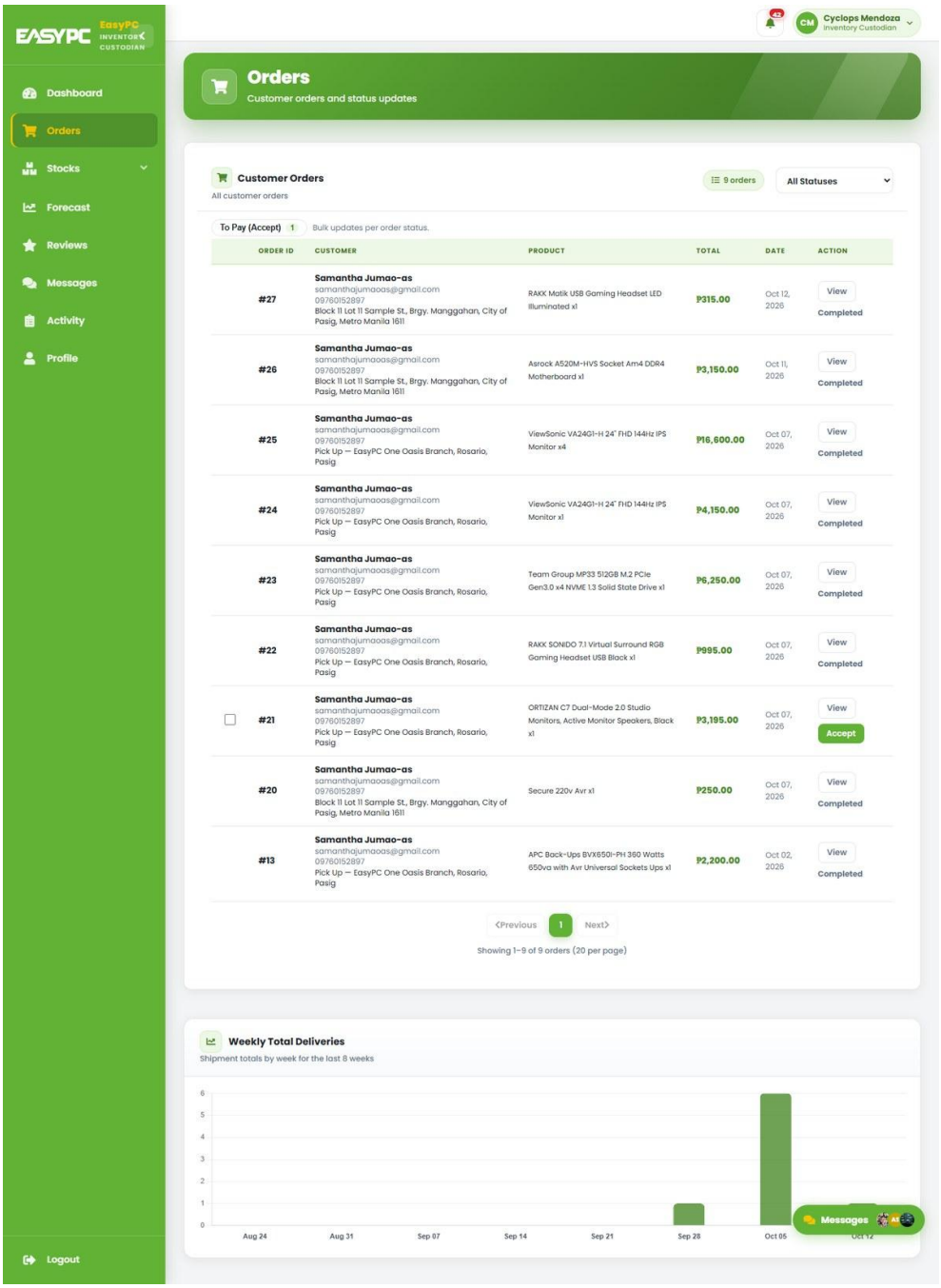


Figure 15. Inventory Custodian: Orders

6.3 Stocks: Inventory

Master list of all products and their stock.

- Search by name, category, SKU or barcode, or scan to search; list or grid view
- Add Product and edit products; status badges (In Stock, Low, Critical, Out of Stock)
- Print barcode labels, or Generate a barcode for items without one

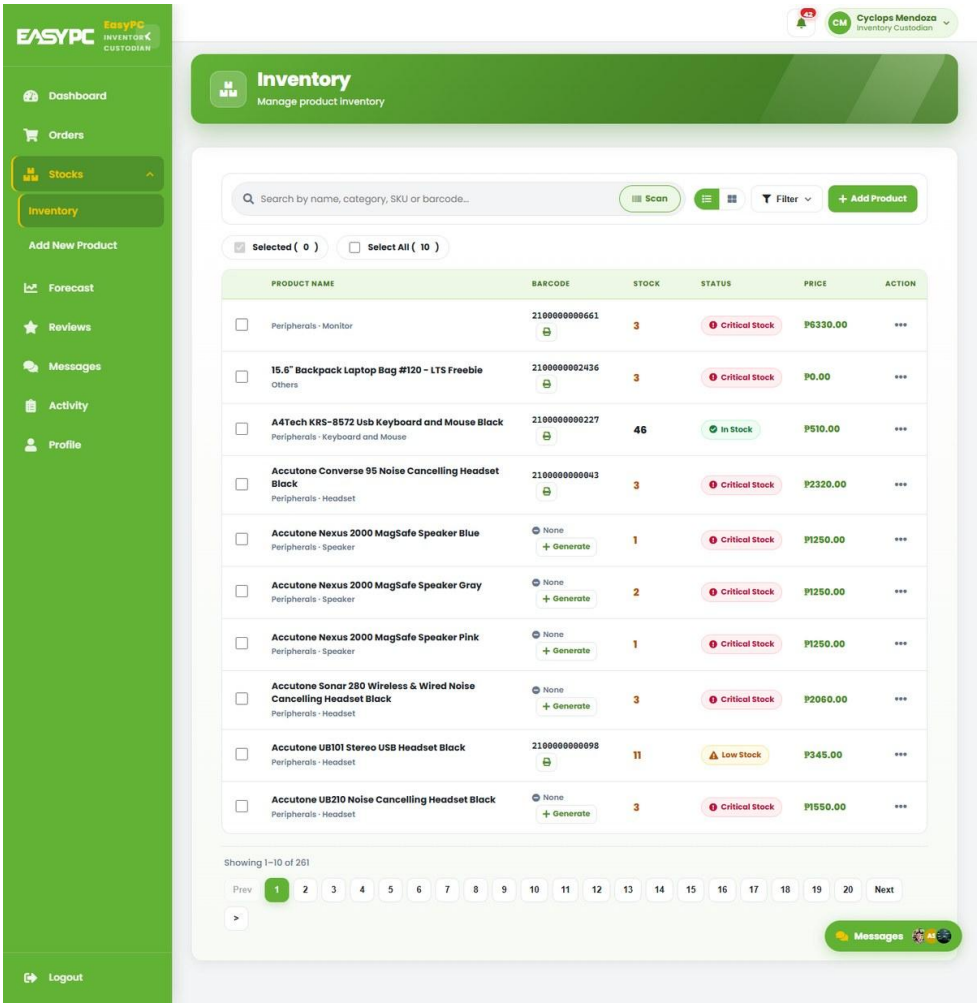


Figure 16. Inventory Custodian: Stocks (Inventory)

6.4 Stocks: Add New Product

Receives deliveries by scanning each item and adding stock to existing products.

- Scan or type a barcode, enter the quantity received, choose the supplier and add an optional delivery reference no.
- Stock updates immediately and is recorded in the activity log
- My Recent Stock-Ins lists the last 10 receiving entries

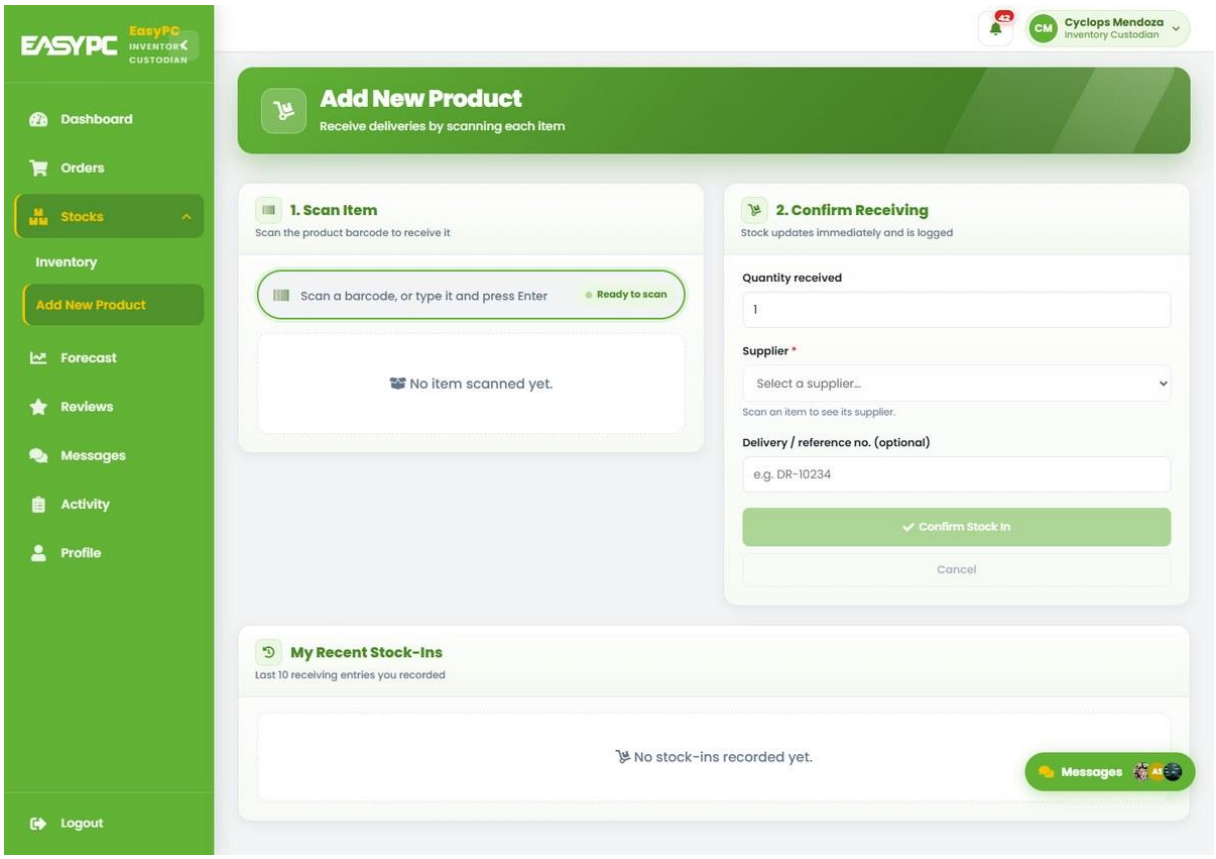


Figure 17. Inventory Custodian: Stocks (Add New Product)

6.5 Forecast

Product Demand Forecast for the next 1–3 months, to support restocking decisions.

- Product name, category, units sold, forecast units, likely range, forecast revenue, trend and confidence
- Filter and table-view controls

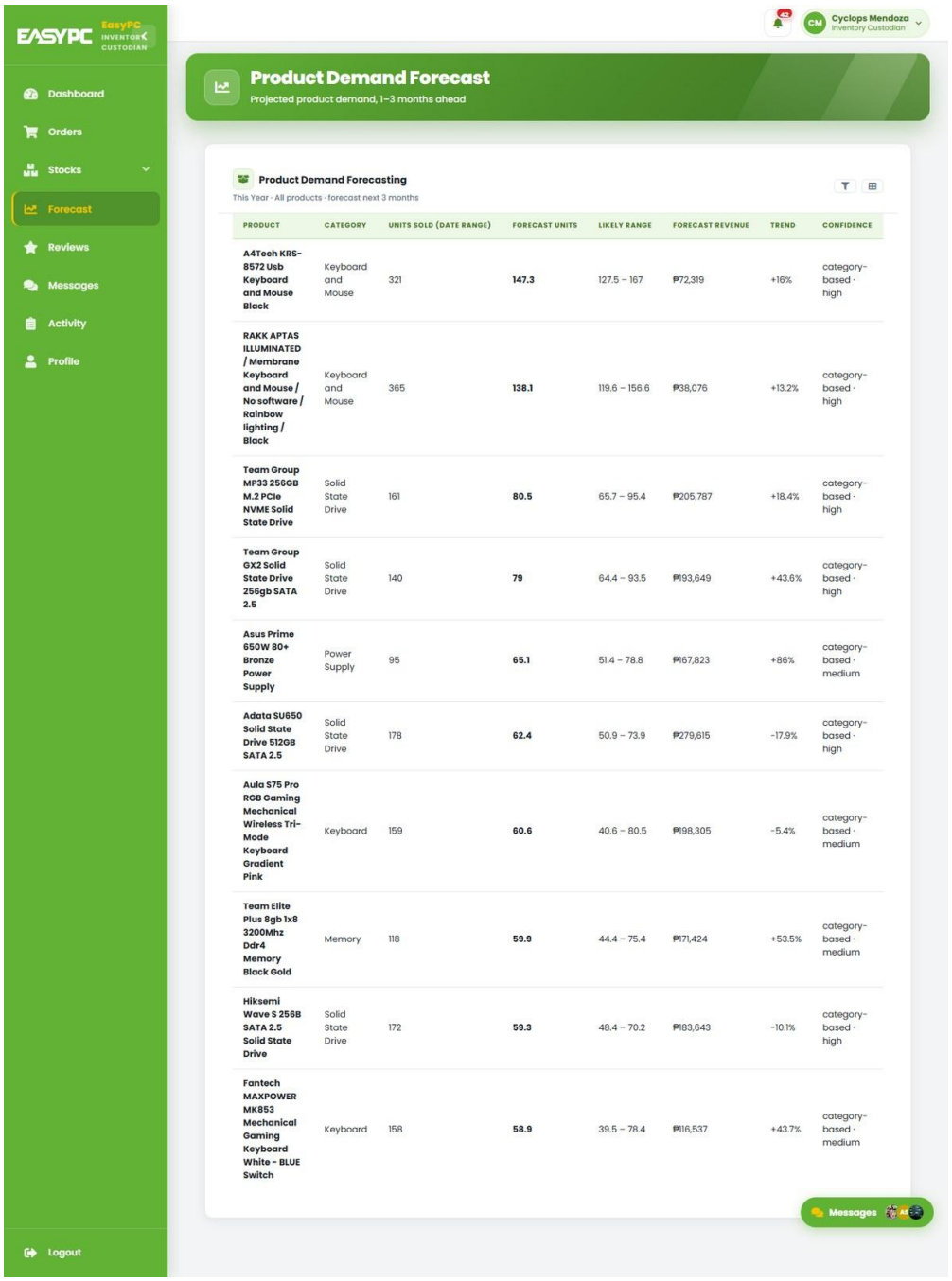


Figure 18. Inventory Custodian: Forecast

6.6 Reviews

Client feedback on purchased products.

- Client, product, order number, review date, star rating and comment (with photos if attached)
- Send a reply, or view the reply already sent
- Filter by star rating or reviews with images; paginated list

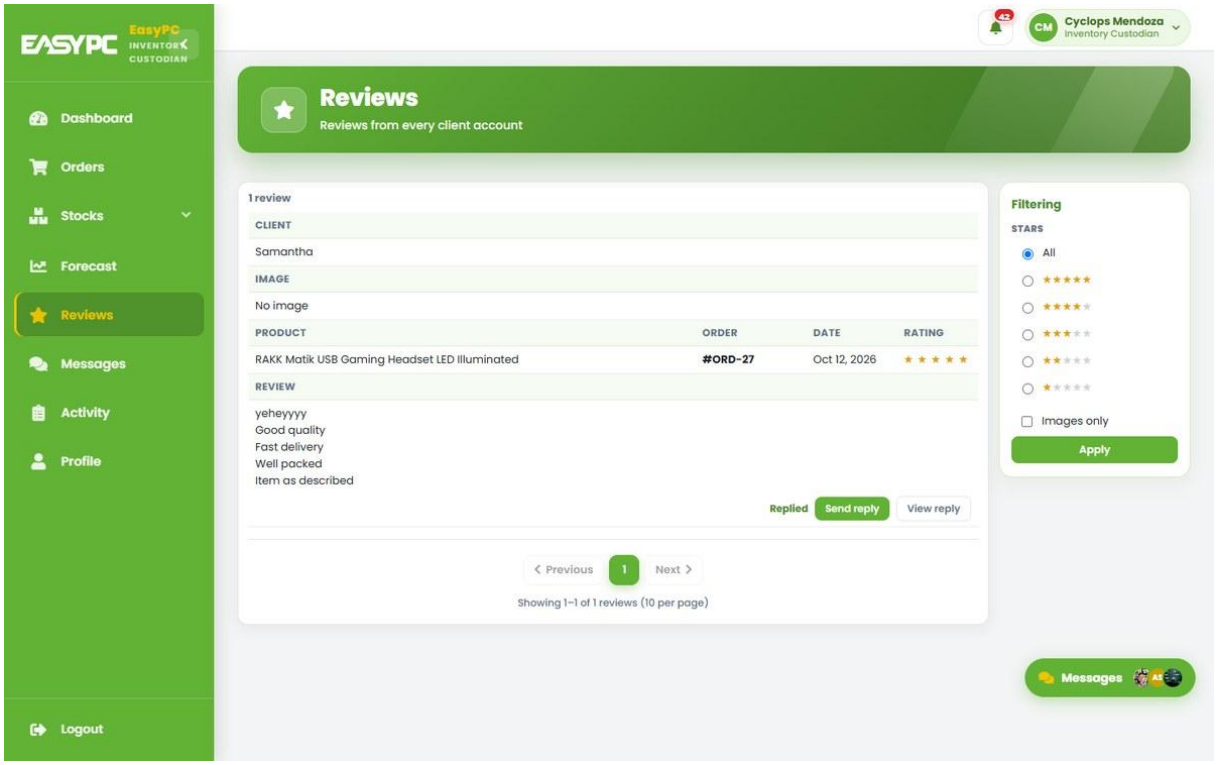


Figure 19. Inventory Custodian: Reviews

6.7 Messages

Internal staff chat with the Admin and Retail Officers.

- Search staff and open a conversation
- Send messages and attachments; react, forward, pin or unsend
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- Also available as a floating Messages widget on every staff page

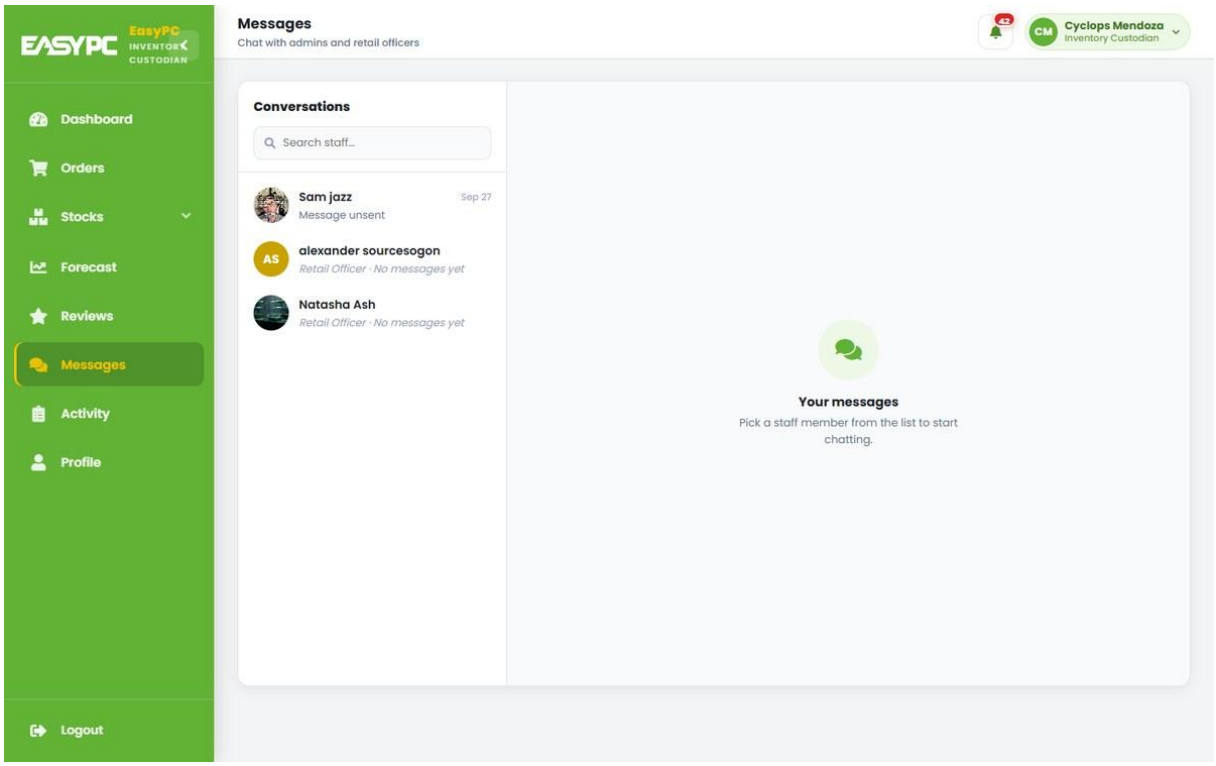


Figure 20. Inventory Custodian: Messages

6.8 Activity

History of inventory changes and custodian actions.

- Order acceptance and updates, stock changes and barcode generation, with user, product and time
- Search by product or user; filter by activity type and period

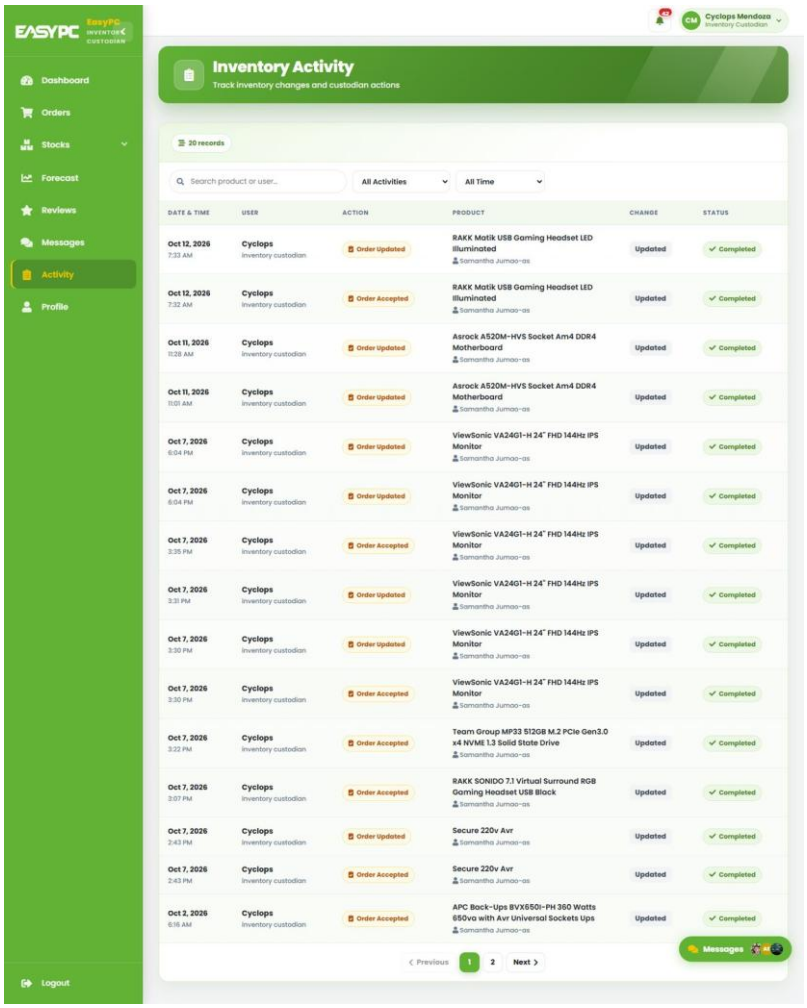


Figure 21. Inventory Custodian: Activity

6.9 Profile

Lets the user update their own account details.

- Edit name, age and address; change the profile picture
- Change password with a live checklist of the password rules

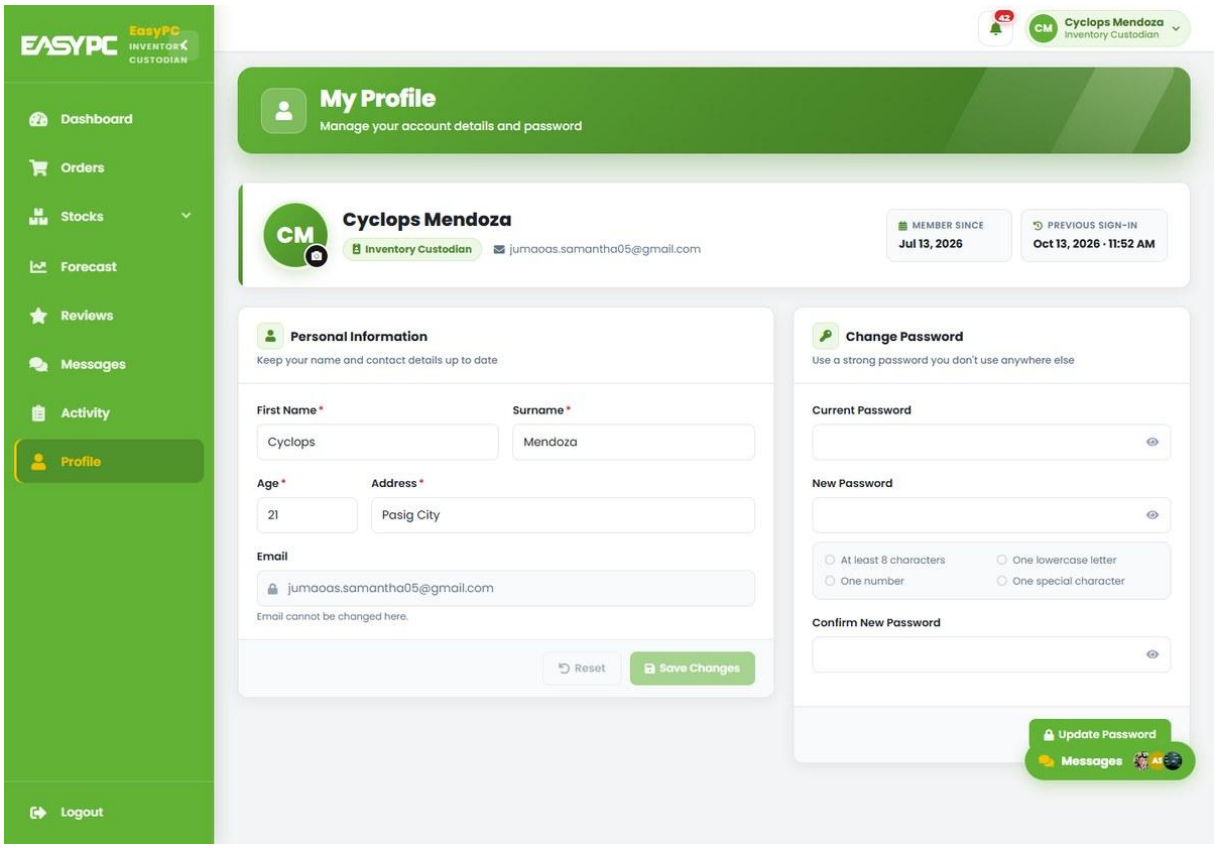


Figure 22. Inventory Custodian: Profile

7. Cashier

The Cashier processes walk-in purchases at the store counter.

7.1 Dashboard

Daily sales summary and stock warnings.

- Today's number of sales and total amount
- Low, critical and out-of-stock counts
- My Recent Transactions with receipt reprint; live Stock Alerts panel
- Start New Sale and View Stock Alerts shortcuts

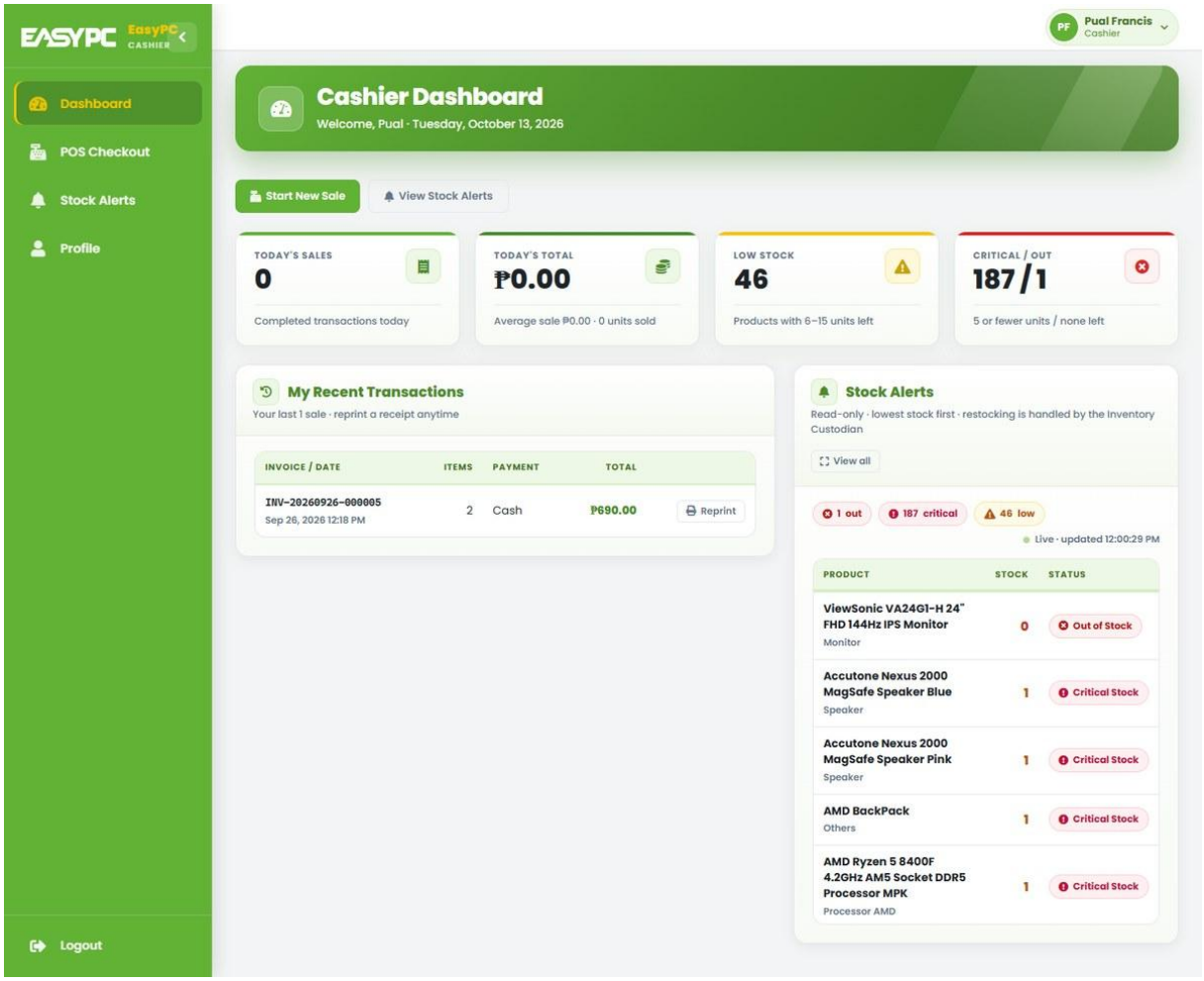


Figure 23. Cashier: Dashboard

7.2 POS Checkout

Scan items to build a sale.

- Scan or type a barcode (UPC-A, UPC-E, EAN-13, EAN-8); scanning again adds one more
- Order Summary computes VATable sales and 12% VAT (prices are VAT-inclusive)
- Pay by Cash, GCash, Maya or Card; stock is deducted automatically

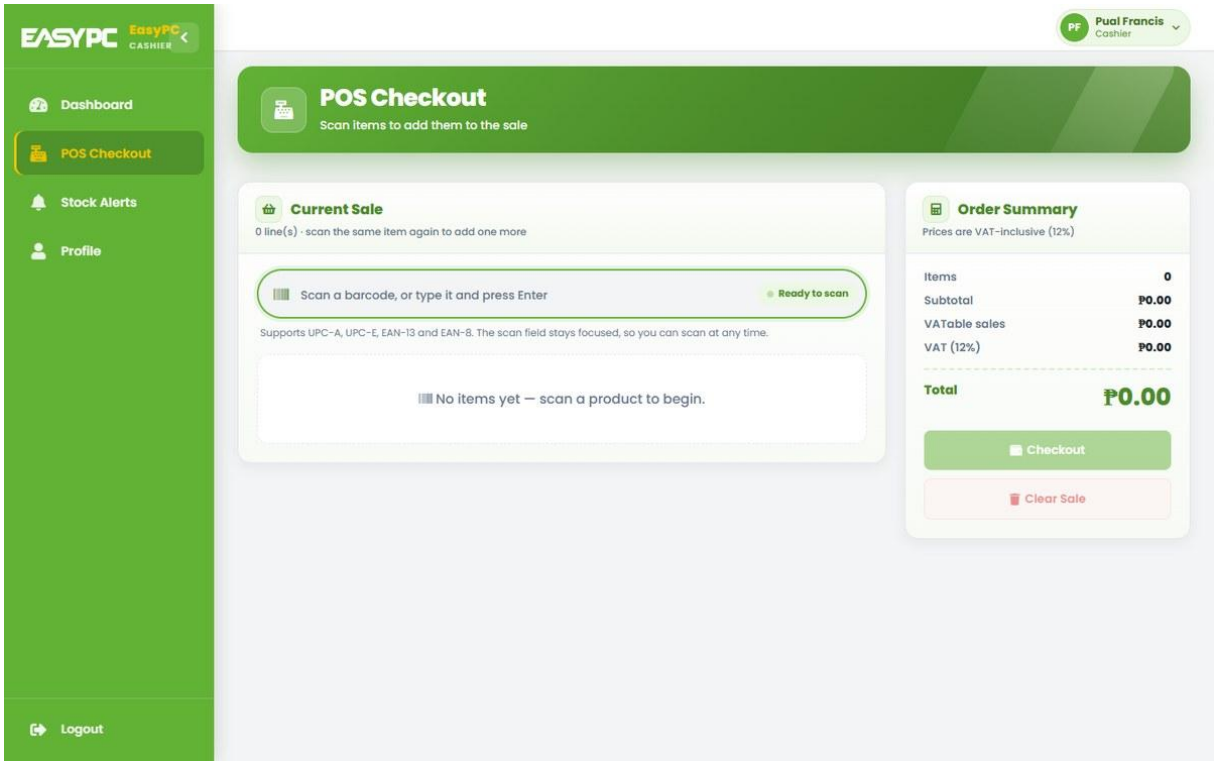


Figure 24. Cashier: POS Checkout

7.3 Receipt

Sales invoice generated after checkout.

- Invoice no., date, cashier, items, VAT breakdown, amount tendered and change
- Print Receipt, start a New Sale or return to the Dashboard

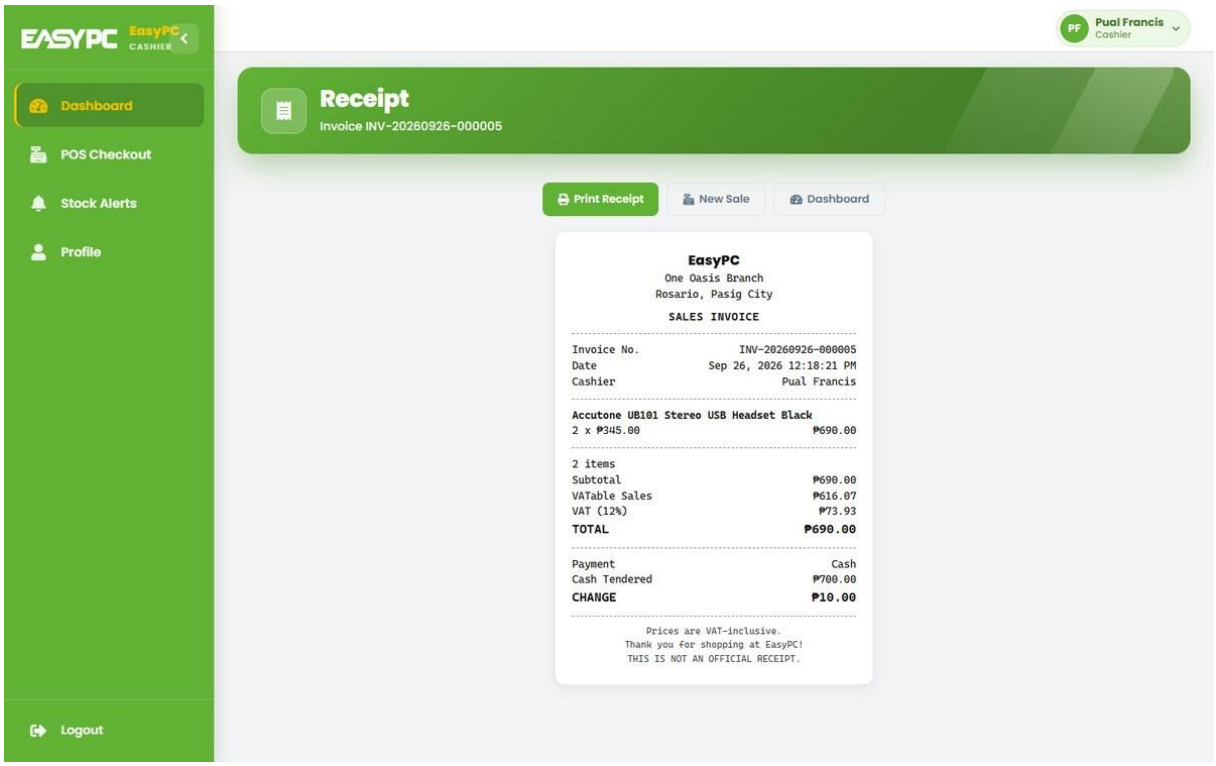


Figure 25. Cashier: Receipt

7.4 Stock Alerts

Read-only list of products that need restocking.

- Counts and list of low (6–15), critical (5 or fewer) and out-of-stock items
- Search, scan and filter; refreshes automatically
- Restocking itself is done by the Inventory Custodian

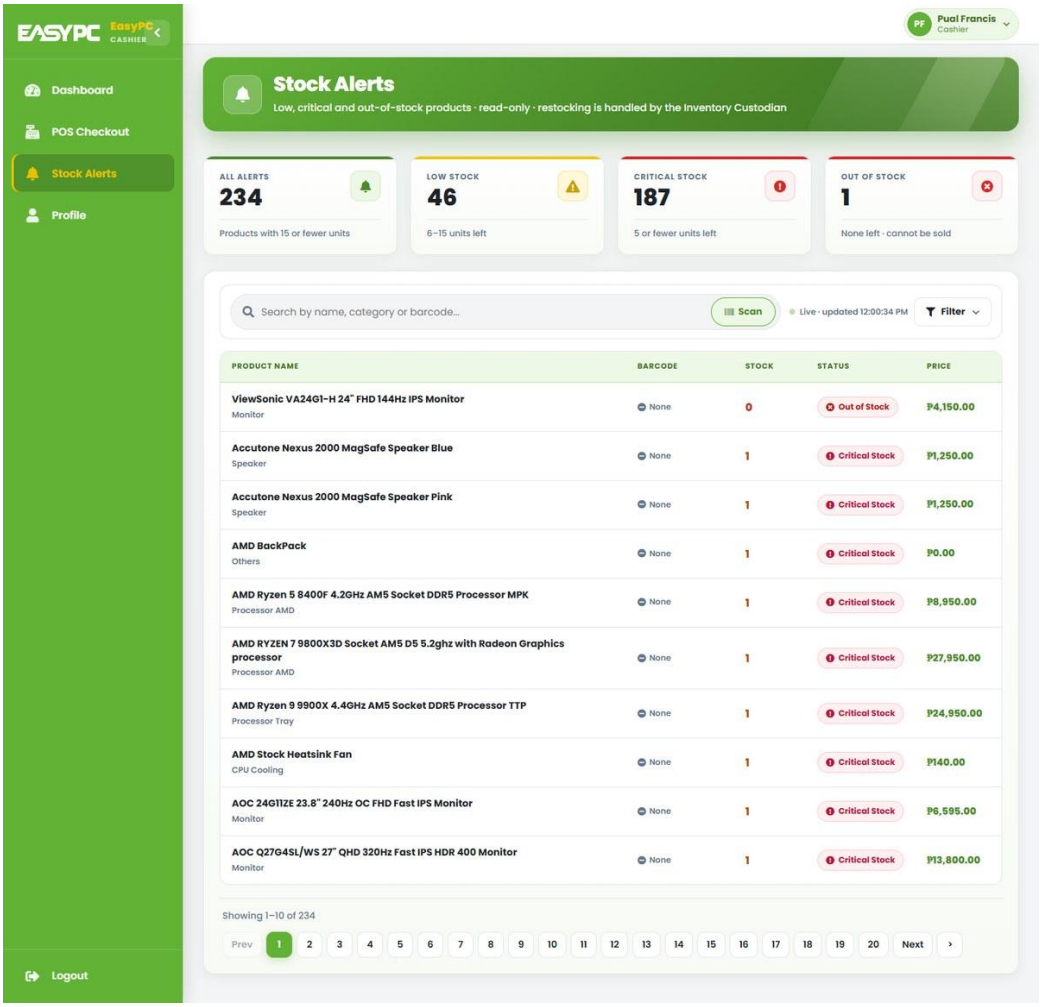


Figure 26. Cashier: Stock Alerts

7.5 Profile

Lets the user update their own account details.

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- Change password with a live checklist of the password rules

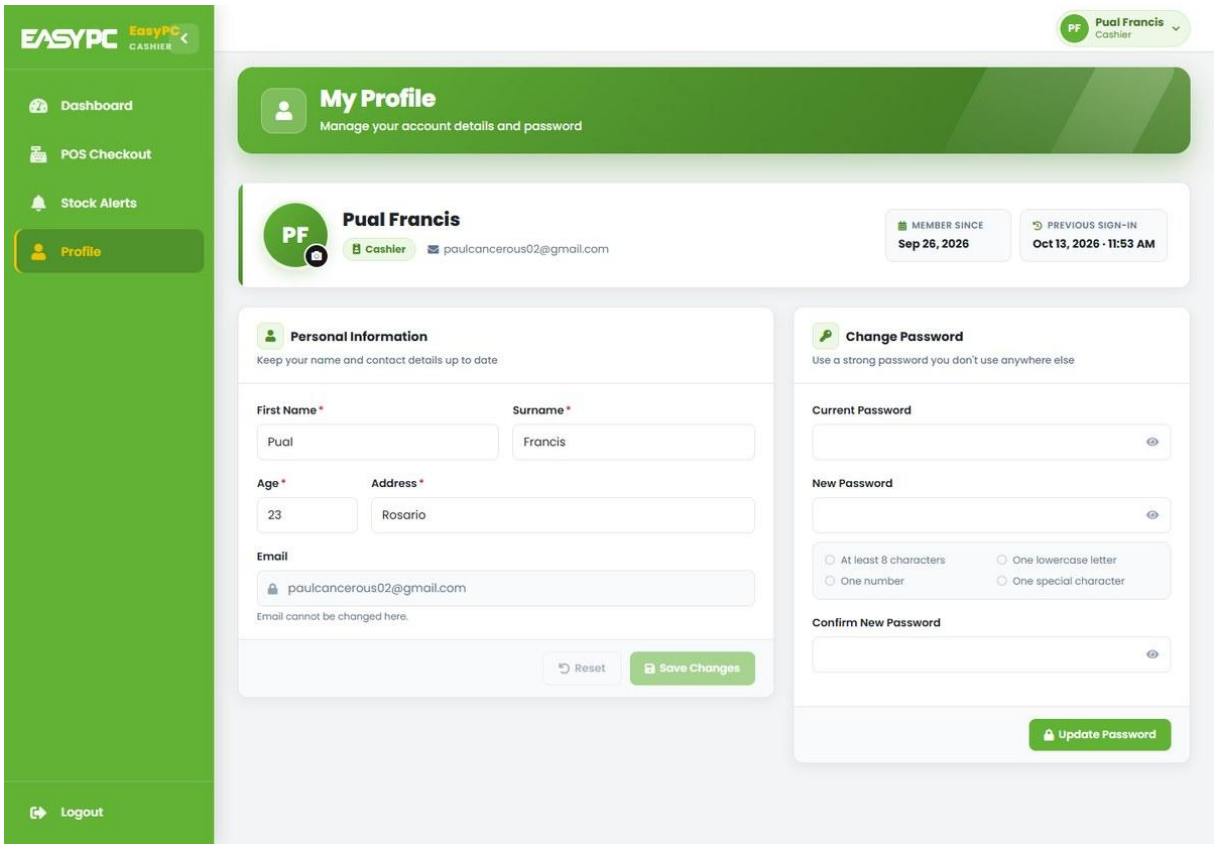


Figure 27. Cashier: Profile

8. Client

Clients browse and buy products online, plan PC builds and get help from Primo.

8.1 Home

Storefront landing page.

- Shop by Category, Featured Products and New Arrivals (items added in the last 7 days)
- Header: product search, notifications, cart and dark-mode toggle
- Footer: store phone number, location and quick links

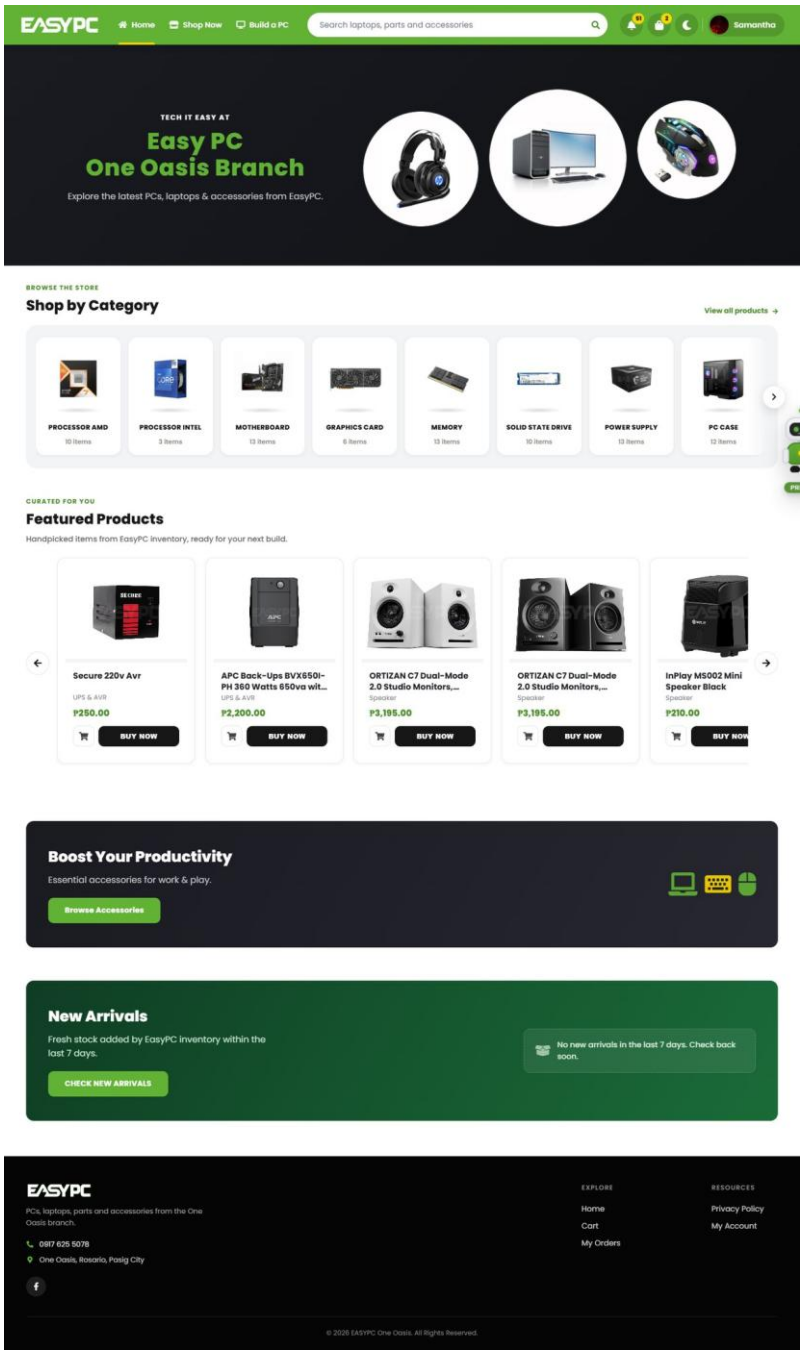


Figure 28. Client: Home

8.2 Primo AI Assistant

Chat assistant available on browsing pages.

- Quick actions: Find a laptop, Track my order, Build a PC, Returns and refunds
- Understands each question by intent and answers from the live catalog, with product cards
- A language model phrases shop-related replies naturally; off-topic questions (e.g. homework, jokes, weather) get a short store-only reply
- Past chats are saved in chat history

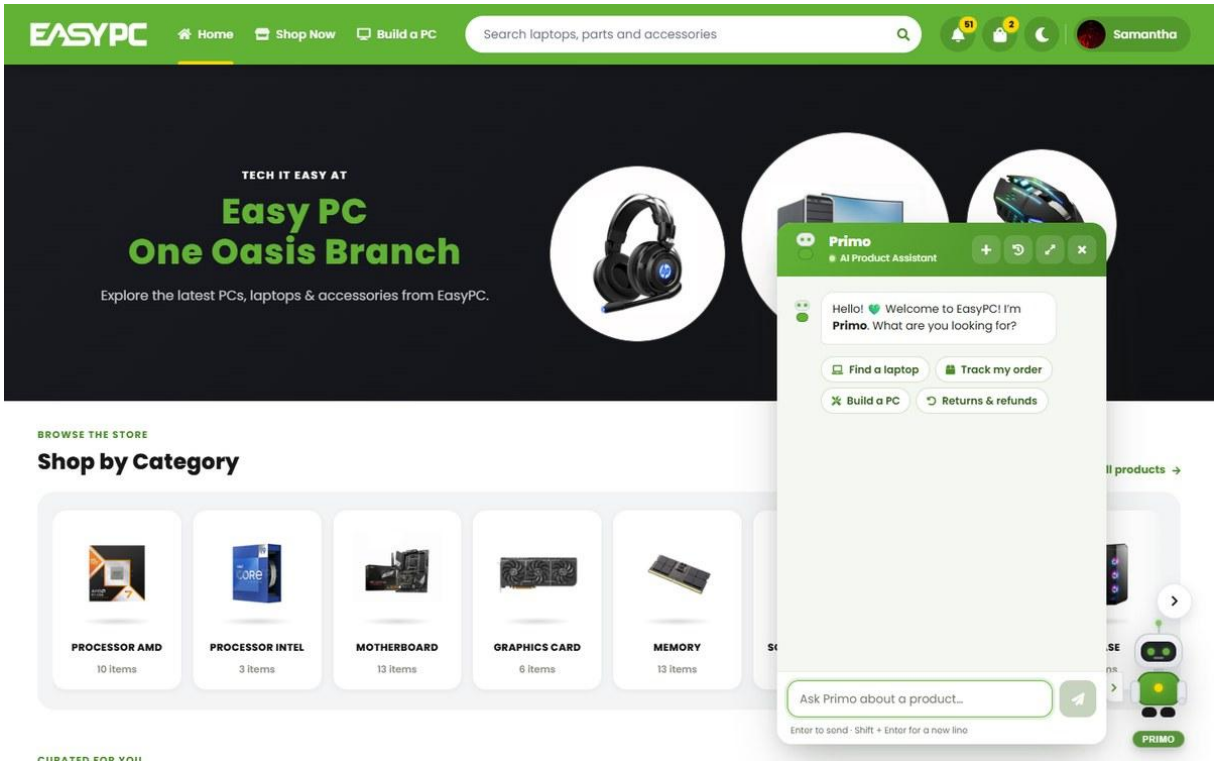


Figure 29. Client: Primo AI Assistant

8.3 Shop

All products in one place.

- Filter by category, price range and brand; quick product groups
- Add to cart, Buy Now or add to wishlist from each card

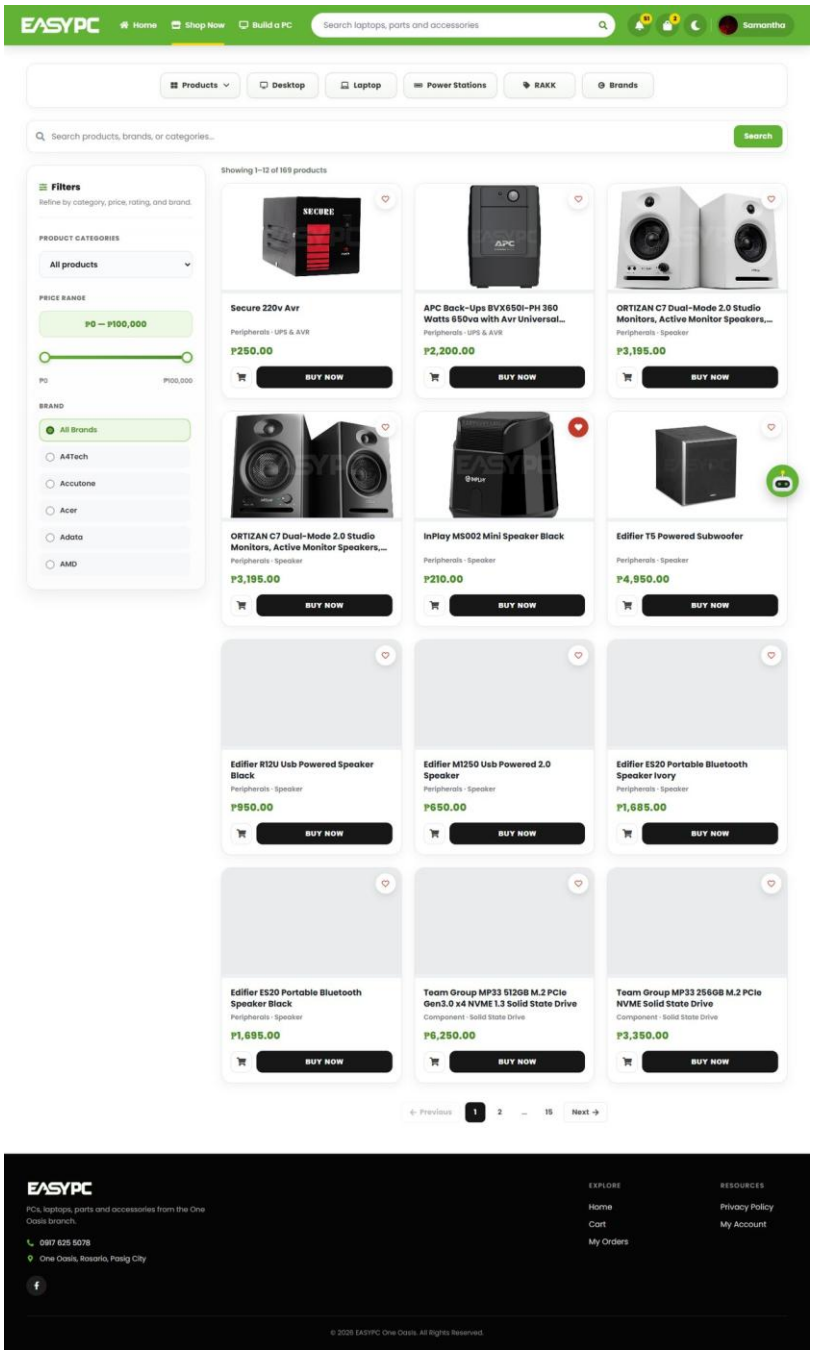


Figure 30. Client: Shop

8.4 Category

Products of one category (e.g. Display).

- Filter by brand, price and availability
- Sort by price (ascending / descending) or rating

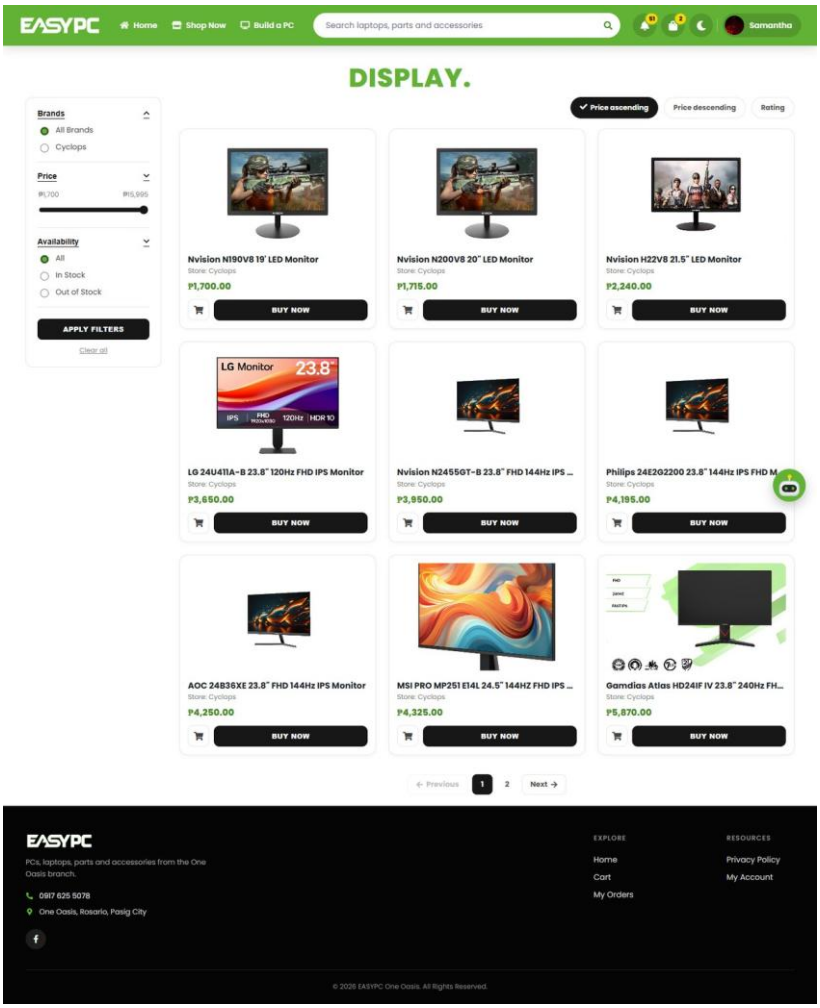


Figure 31. Client: Category

8.5 Product Details

Full information on one product.

- Image zoom, price, stock left and quantity selector
- Add to Cart, Buy Now and Add to Wishlist
- Overview, Specifications, Reviews and Related Products

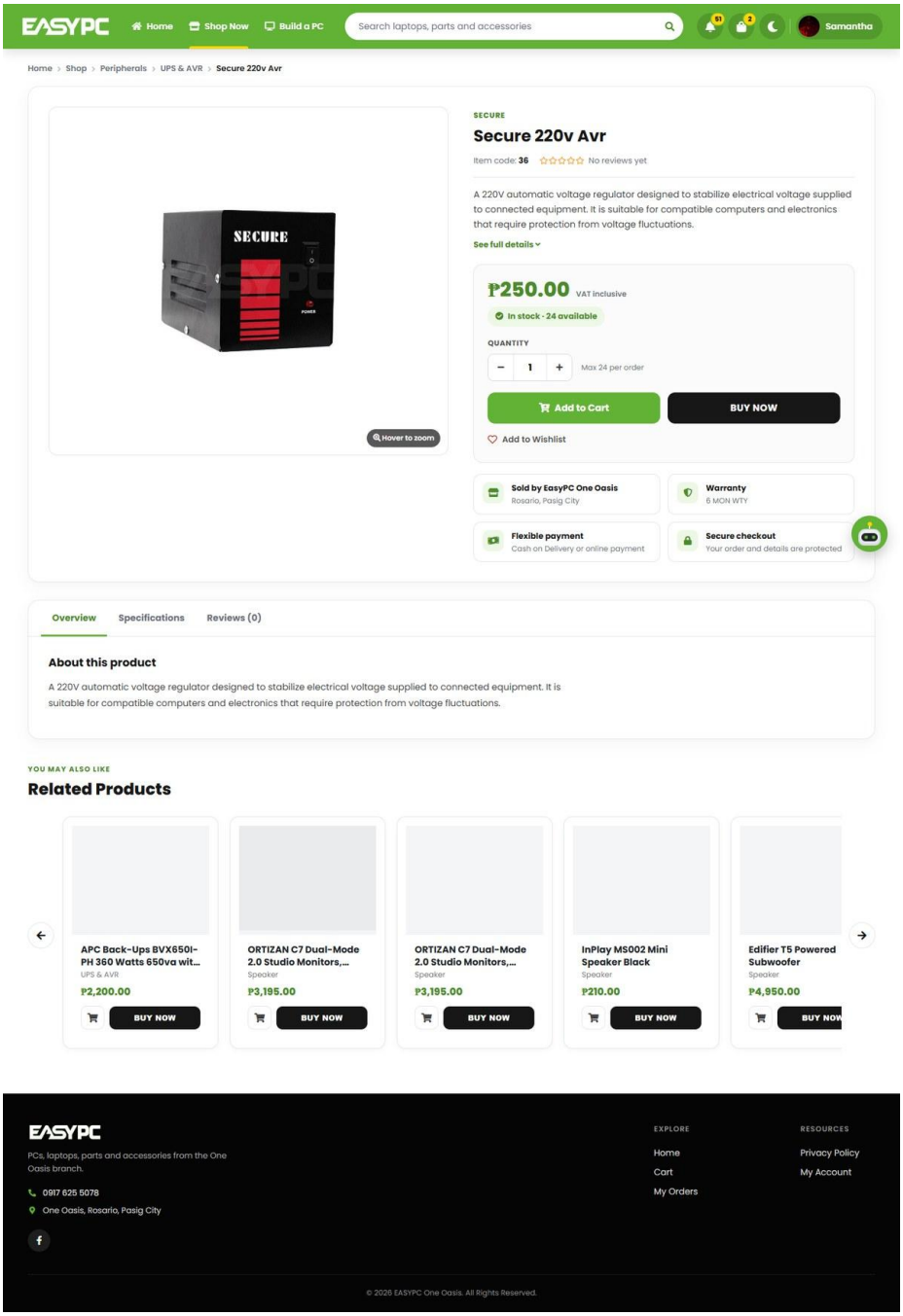


Figure 32. Client: Product Details

8.6 Build a PC

Step-by-step PC builder.

- Choose parts by step: Core, Memory, Graphics, Storage, Power, Case, Cooling
- 3D / 2D case view, build score, performance balance and power estimate
- Compatibility checks; save, share, load and compare builds

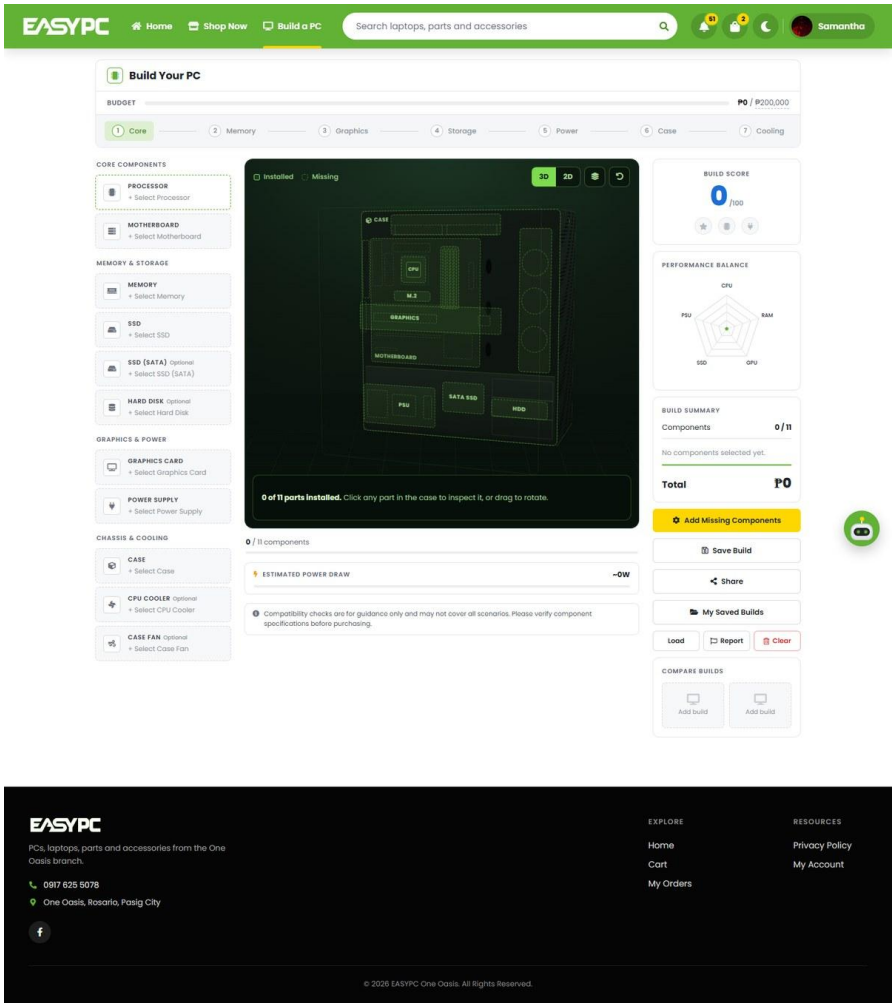


Figure 33. Client: Build a PC

8.7 Saved Builds

PC builds saved by the client.

- Shows parts, total price and whether the build is complete
- Add the whole build to cart, view, edit or delete; search and sort

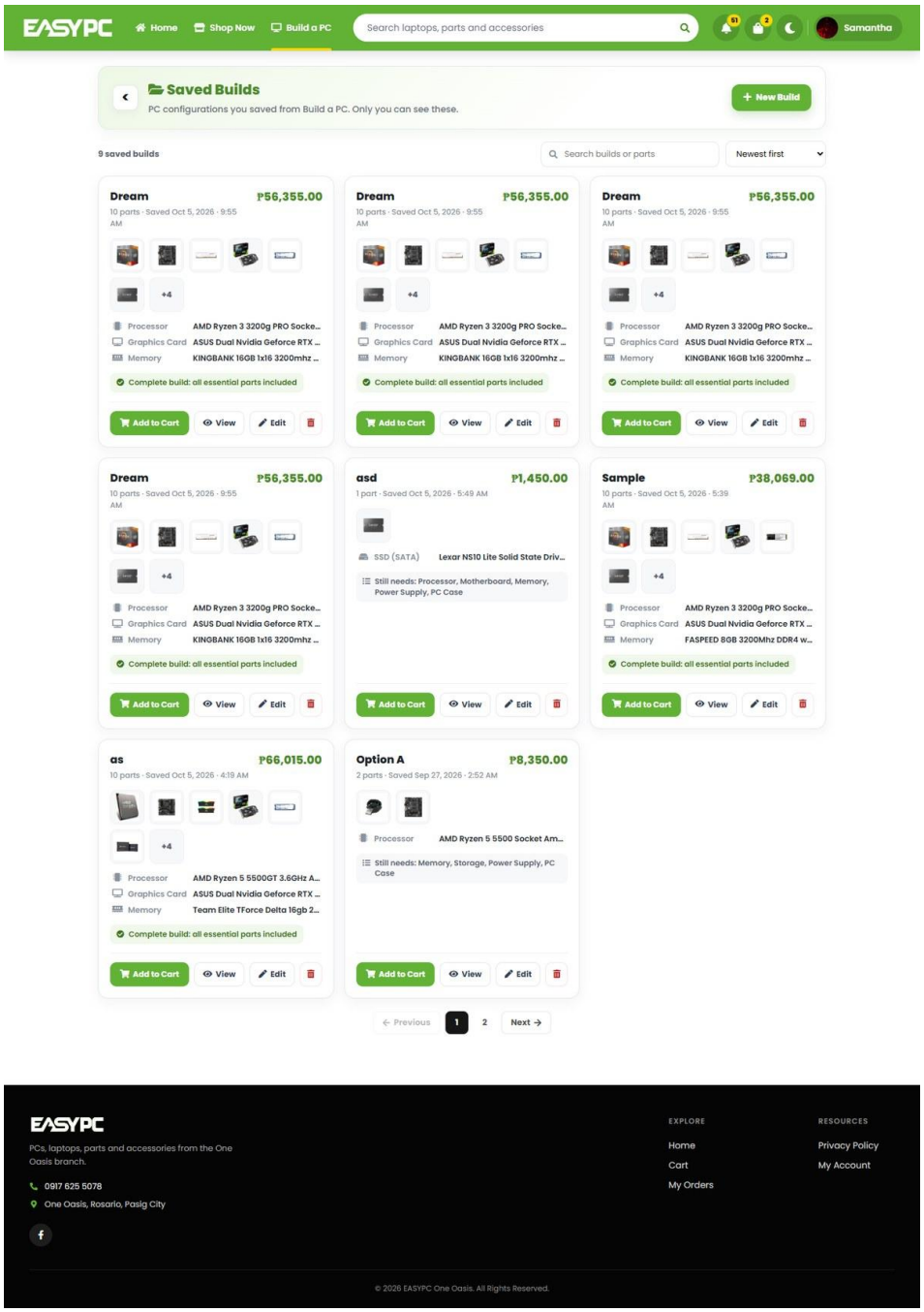


Figure 34. Client: Saved Builds

8.8 Wishlist

Products saved for later.

- Shows price and stock; Buy Now or remove from the list

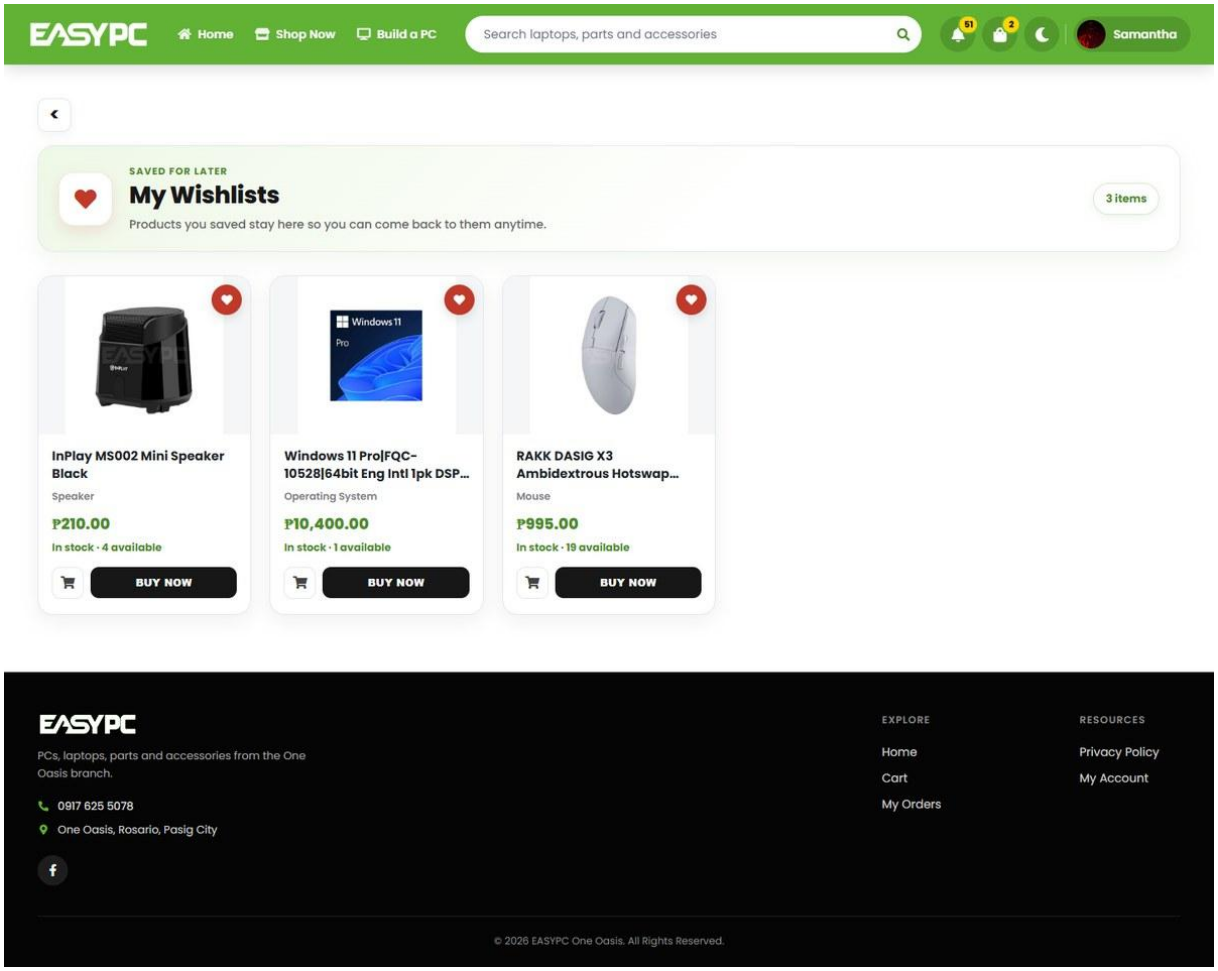


Figure 35. Client: Wishlist

8.9 Cart

Items the client plans to buy.

- Select items, change quantity or remove
- Order Summary with total, then proceed to Checkout

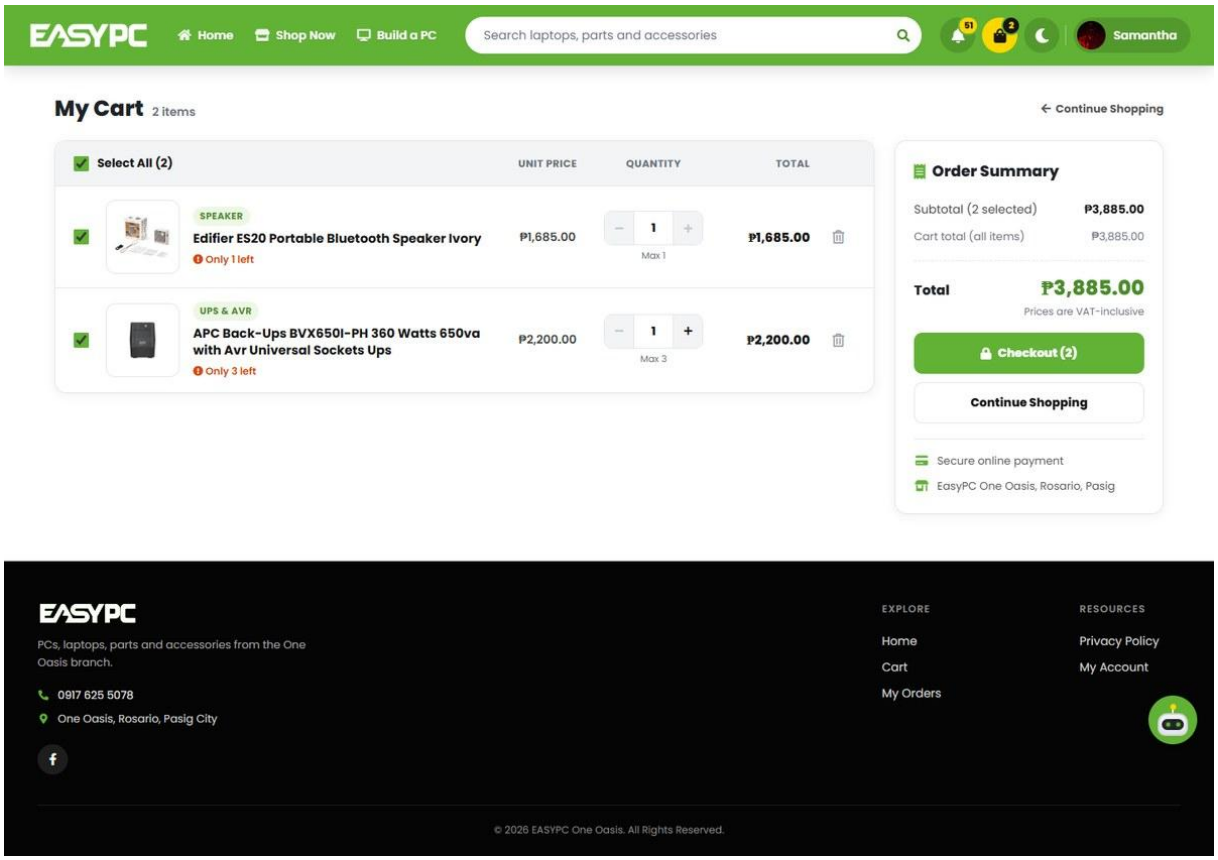


Figure 36. Client: Cart

8.10 Checkout

Confirms shipping details and places the order.

- Order method: saved address, a different address, or Pick Up at the store
- Delivery is through Lalamove; payment is made afterwards under To Pay

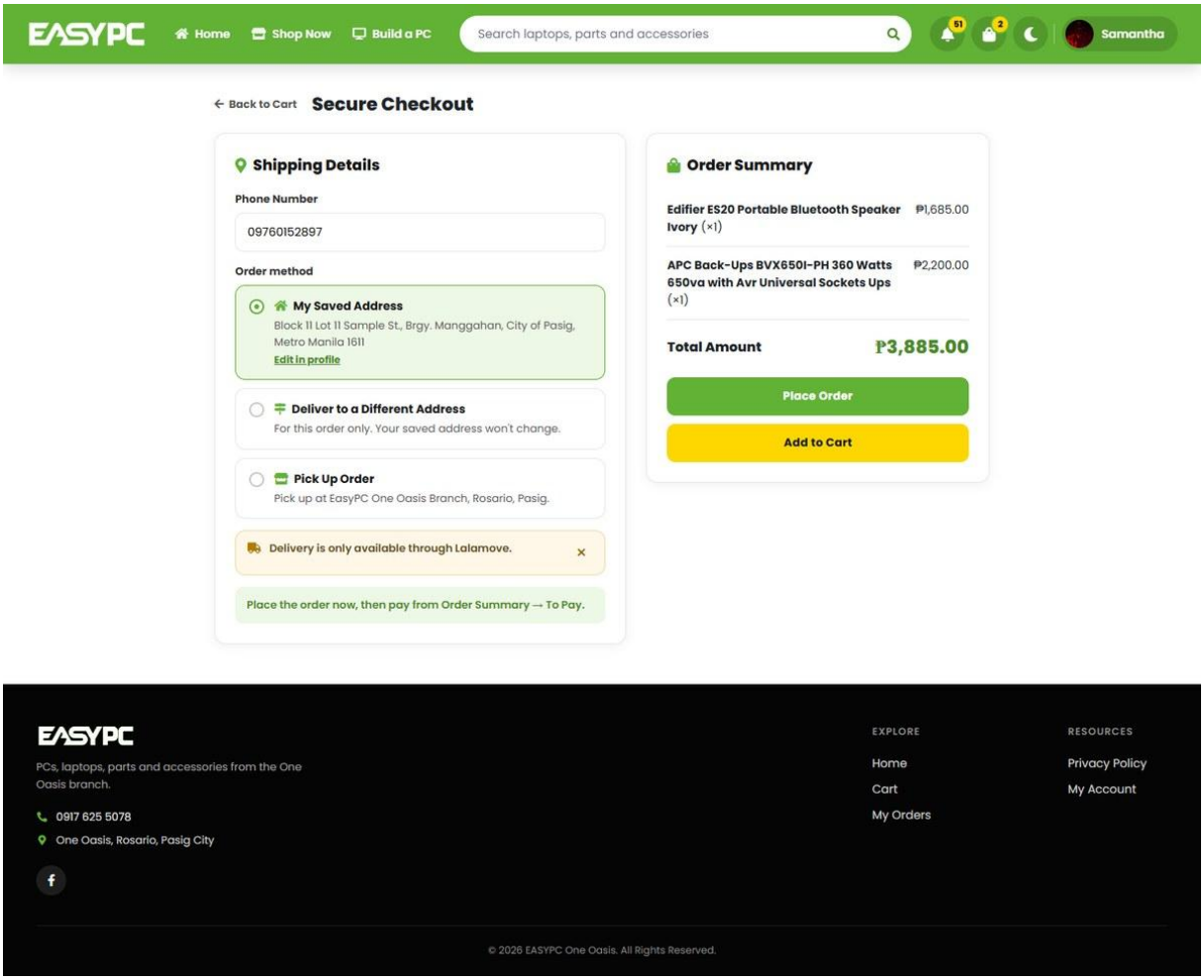


Figure 37. Client: Checkout

8.11 My Account and Orders

Client's account hub.

- Order tabs: All, To pay, To ship, To receive; plus Order History (received and cancelled orders)
- Order details and Your review for received orders
- Delivery address, wishlist, saved builds and notification shortcuts; New in store

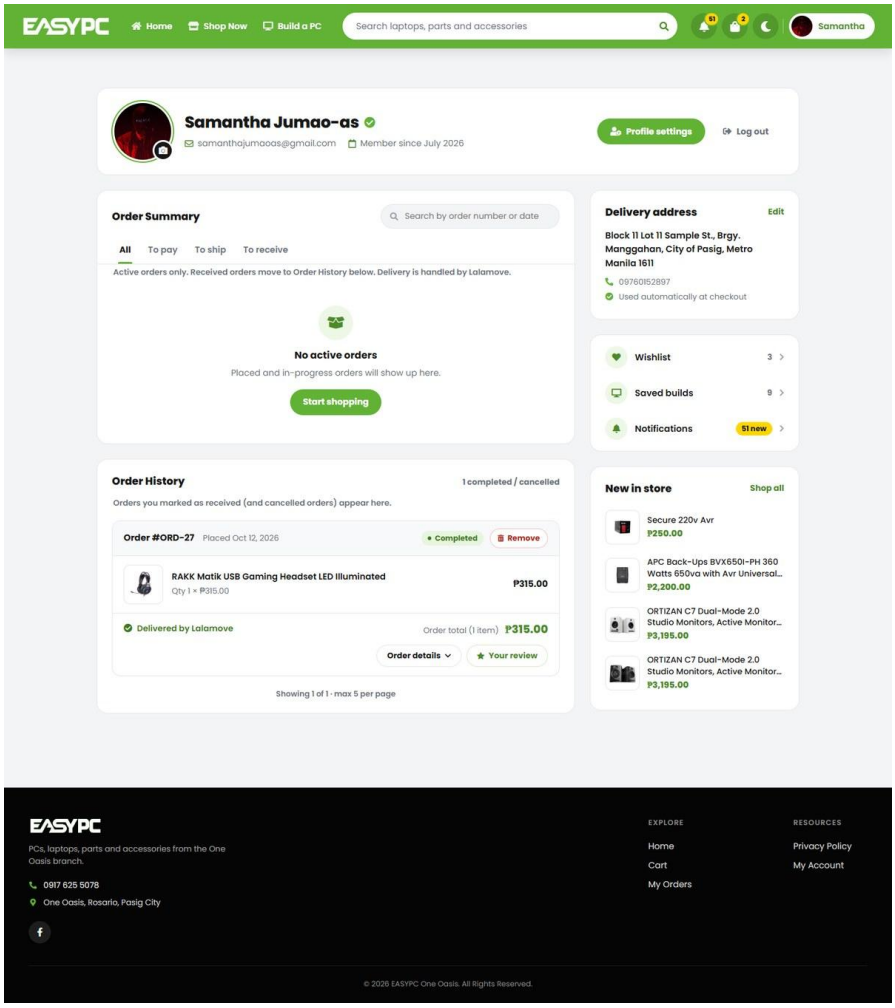


Figure 38. Client: My Account and Orders

8.12 Product Review

Lets clients rate products after receiving their orders.

- 1–5 star rating with a written comment
- Option to post the review without showing their name
- Shows the store's reply to the review

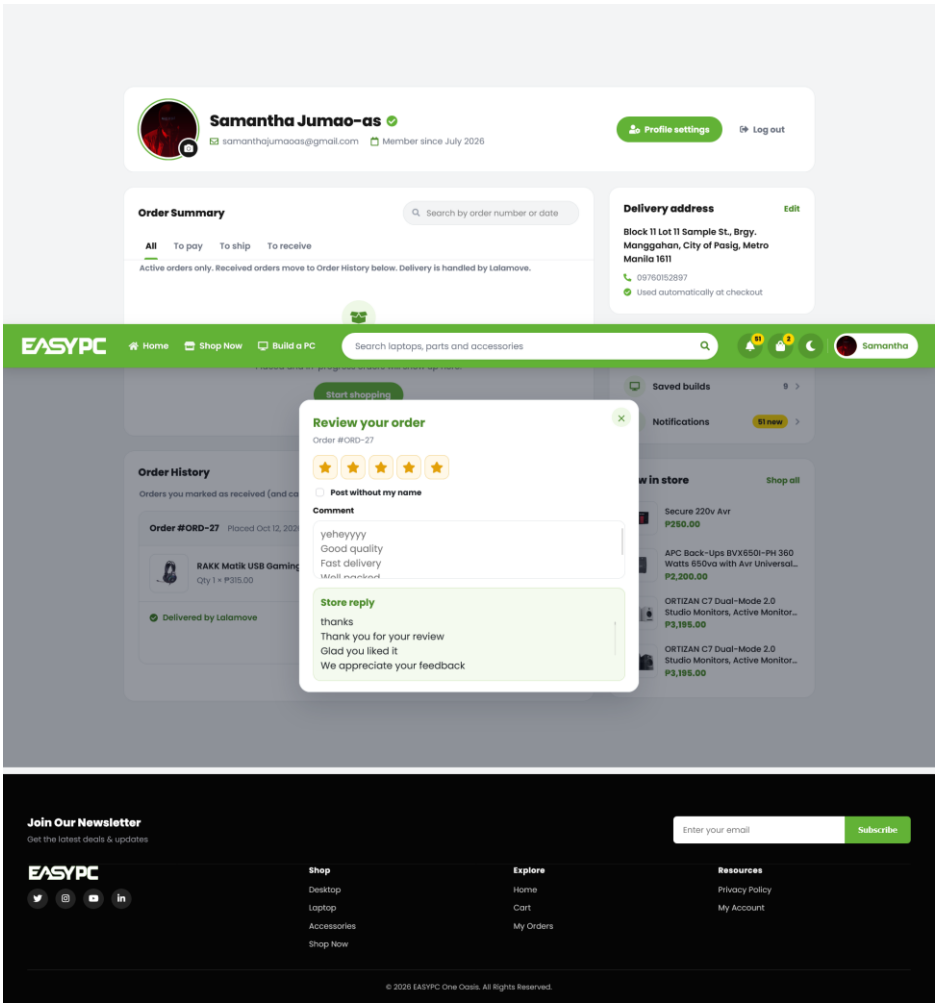


Figure 39. Client: Product Review

8.13 Account Settings

Edit personal and delivery details.

- Name, age, email and phone number
- Default delivery address (province, city, barangay, ZIP) used at checkout

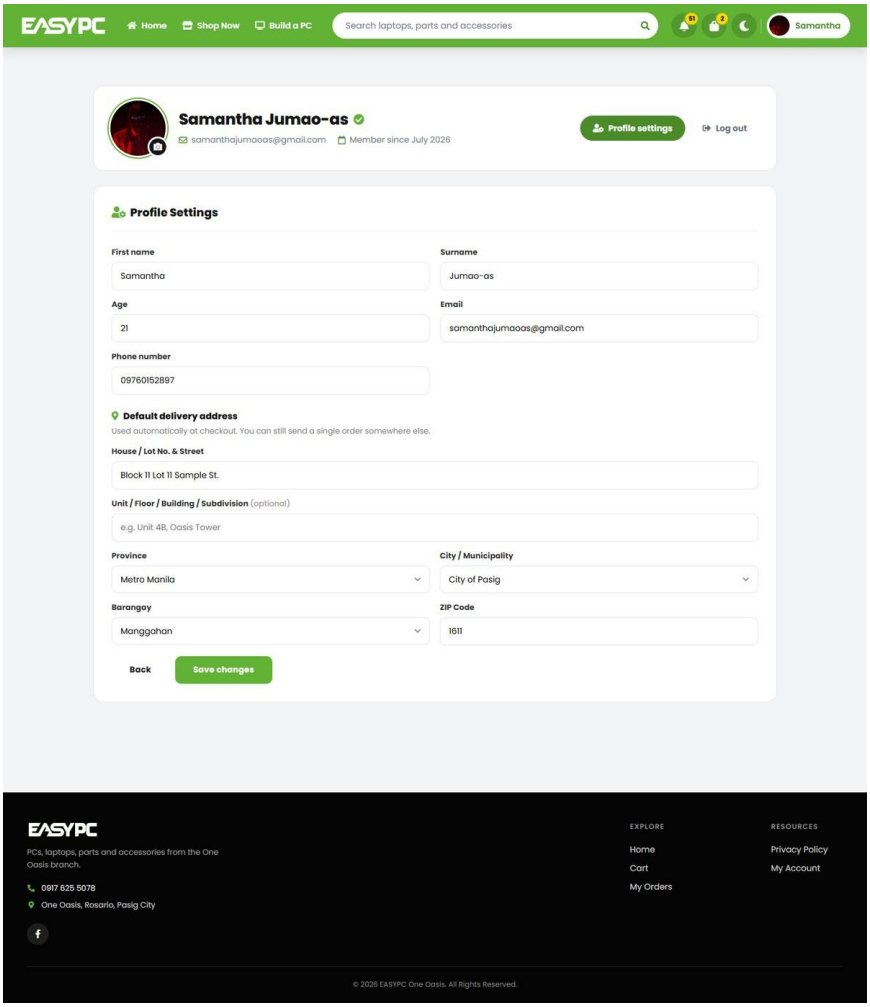


Figure 40. Client: Account Settings

8.14 Notifications

Updates about orders and payments.

- Order, payment and cancellation updates with links to the order
- Filter by type; mark one or all as read; page through older notifications

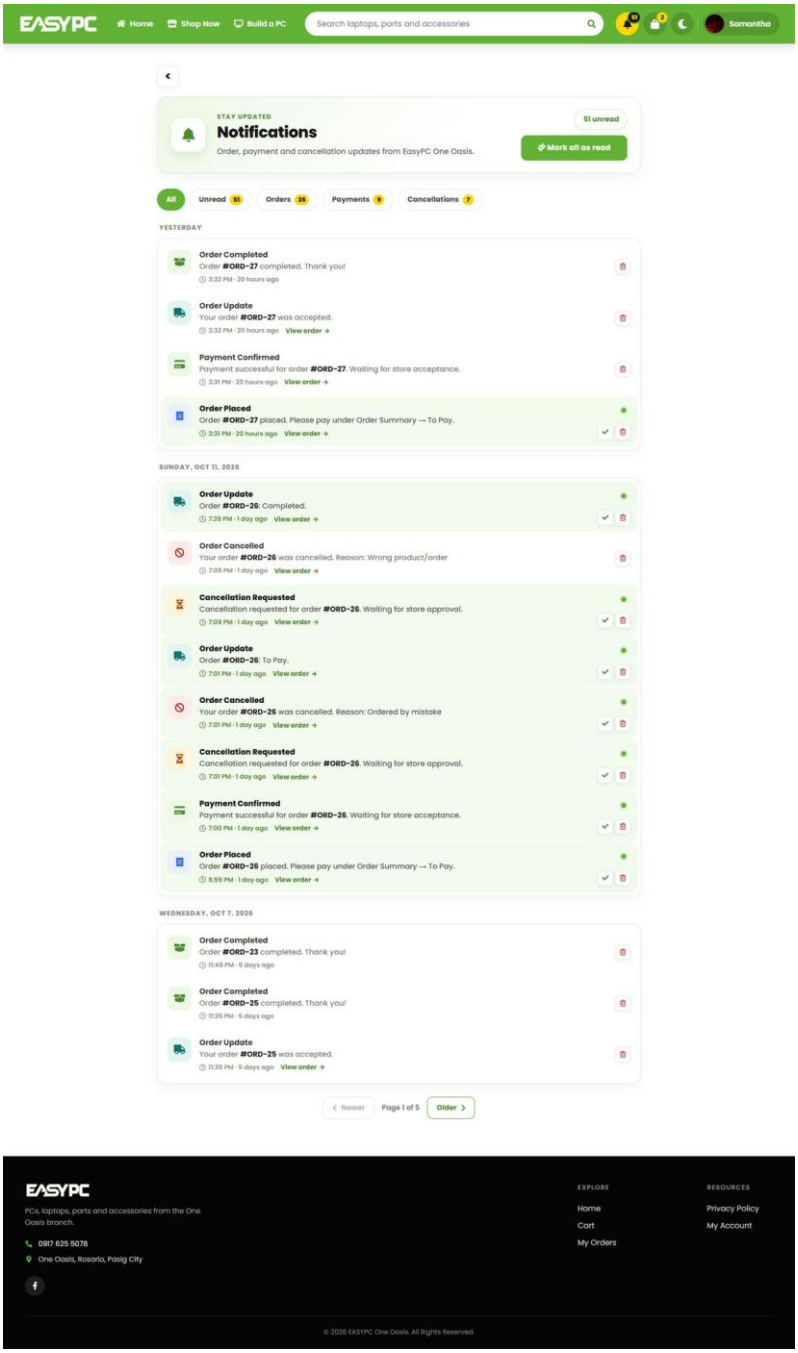


Figure 41. Client: Notifications

8.15 Privacy Policy

How the store handles customer data.

- Information collected and how it is used, payments, the Primo assistant, who can see your information, security measures, cookies, data retention, your rights under the Data Privacy Act of 2012 and policy changes
- “On this page” section list and a contact card to call or message the store

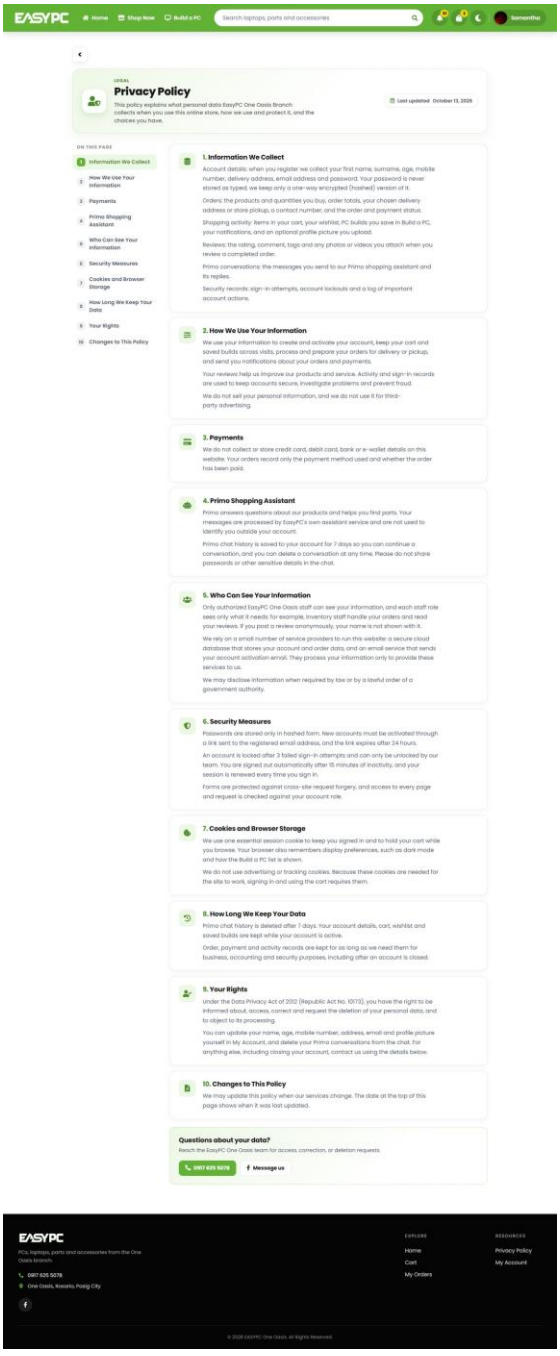


Figure 42. Client: Privacy Policy